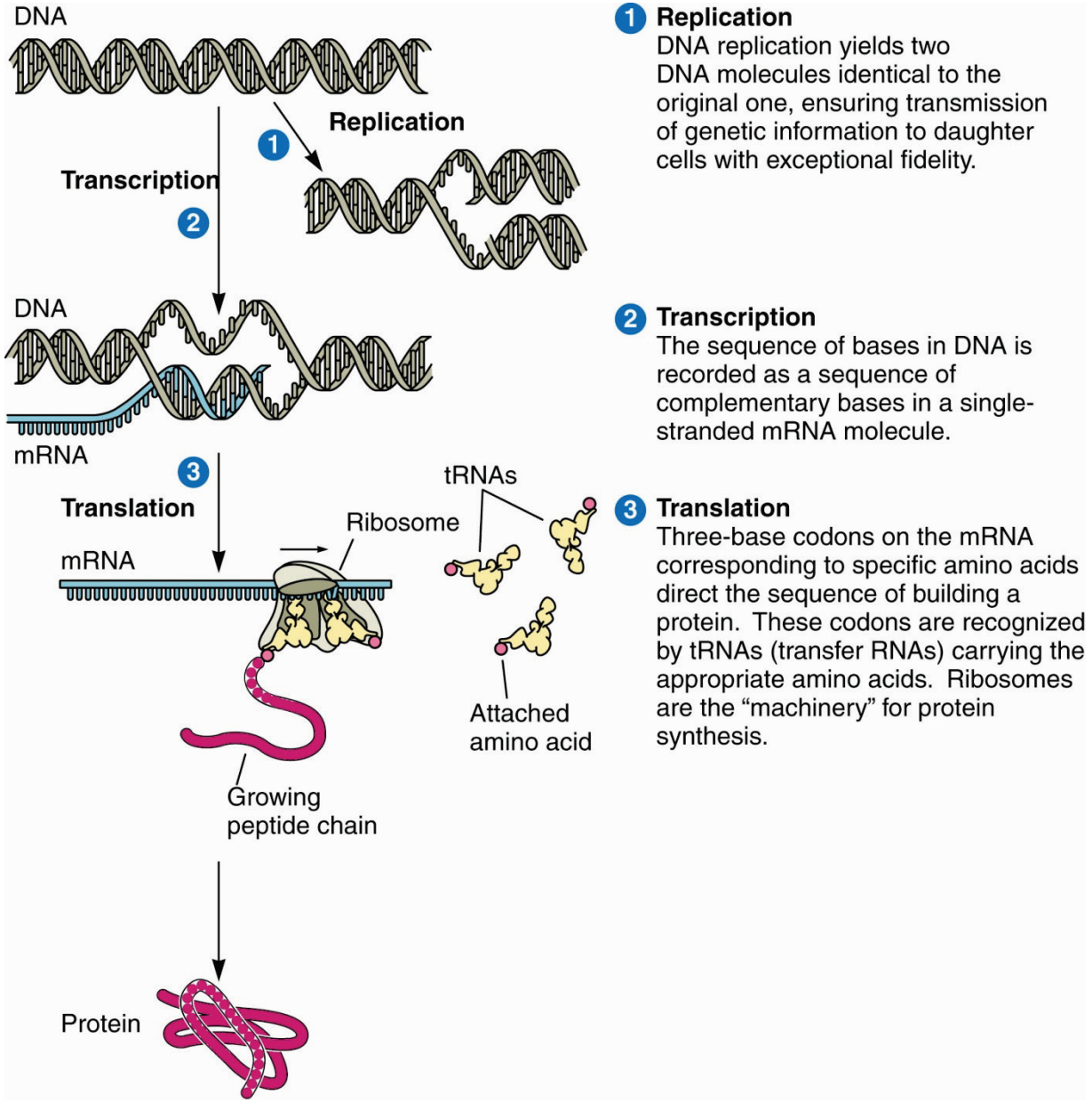
Nucleotides and nucleic acids I and II

Dr. Matias Avila
maavila@unav.es

Nucleic acids are at the center of the “central dogma” of molecular biology



Friedrich Miescher

(Basel, Switzerland 1844-1895)



- Performed the first systematic analyses of the cell's nucleus.

- In 1868 isolated from cells' nuclei a substance that contained phosphorous and that he called "nuclein".

- He demonstrated that nuclein contained an acidic part (DNA) and a basic part (proteins).

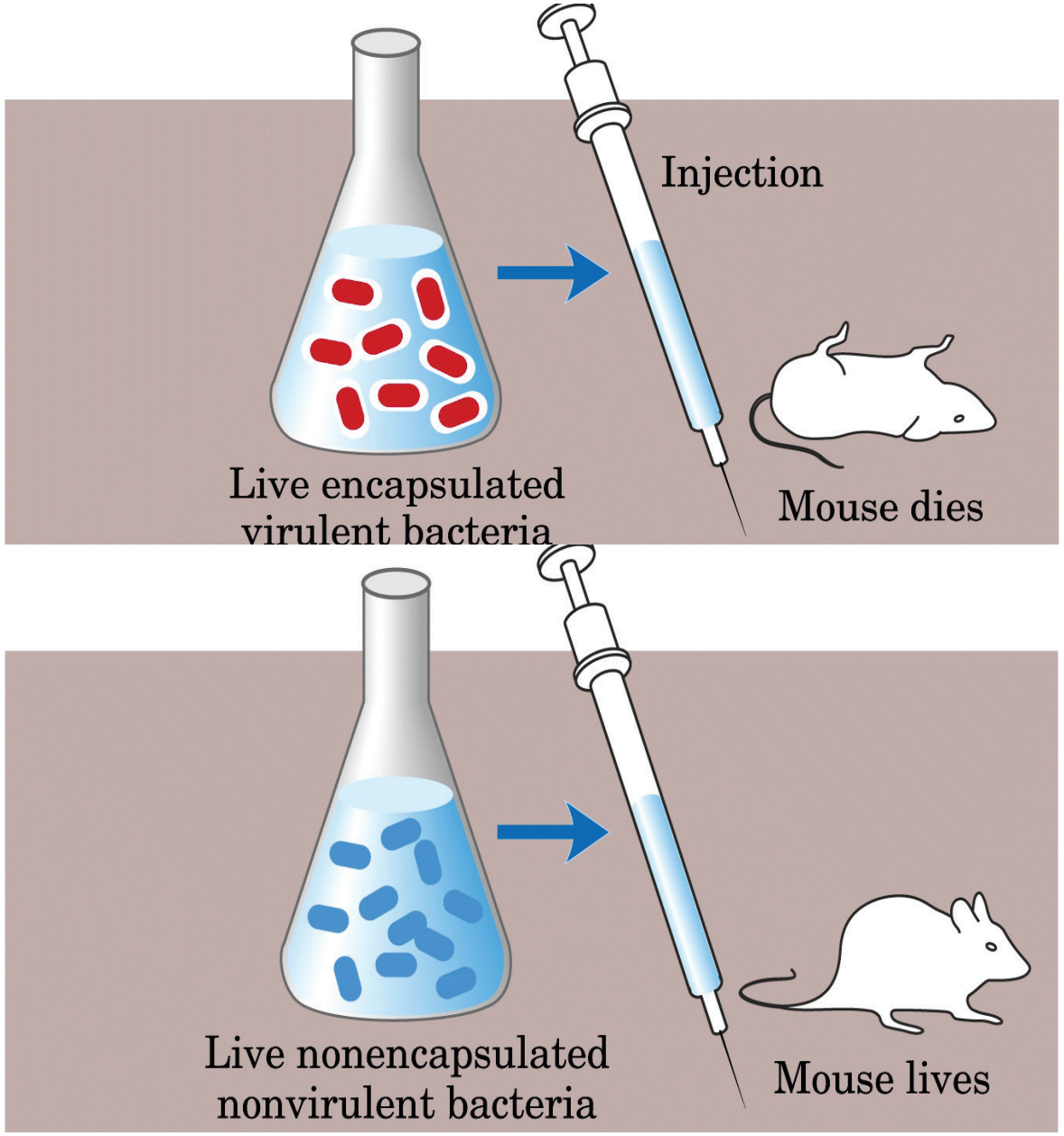
- He proposed that nuclein must be related somehow to heredity.

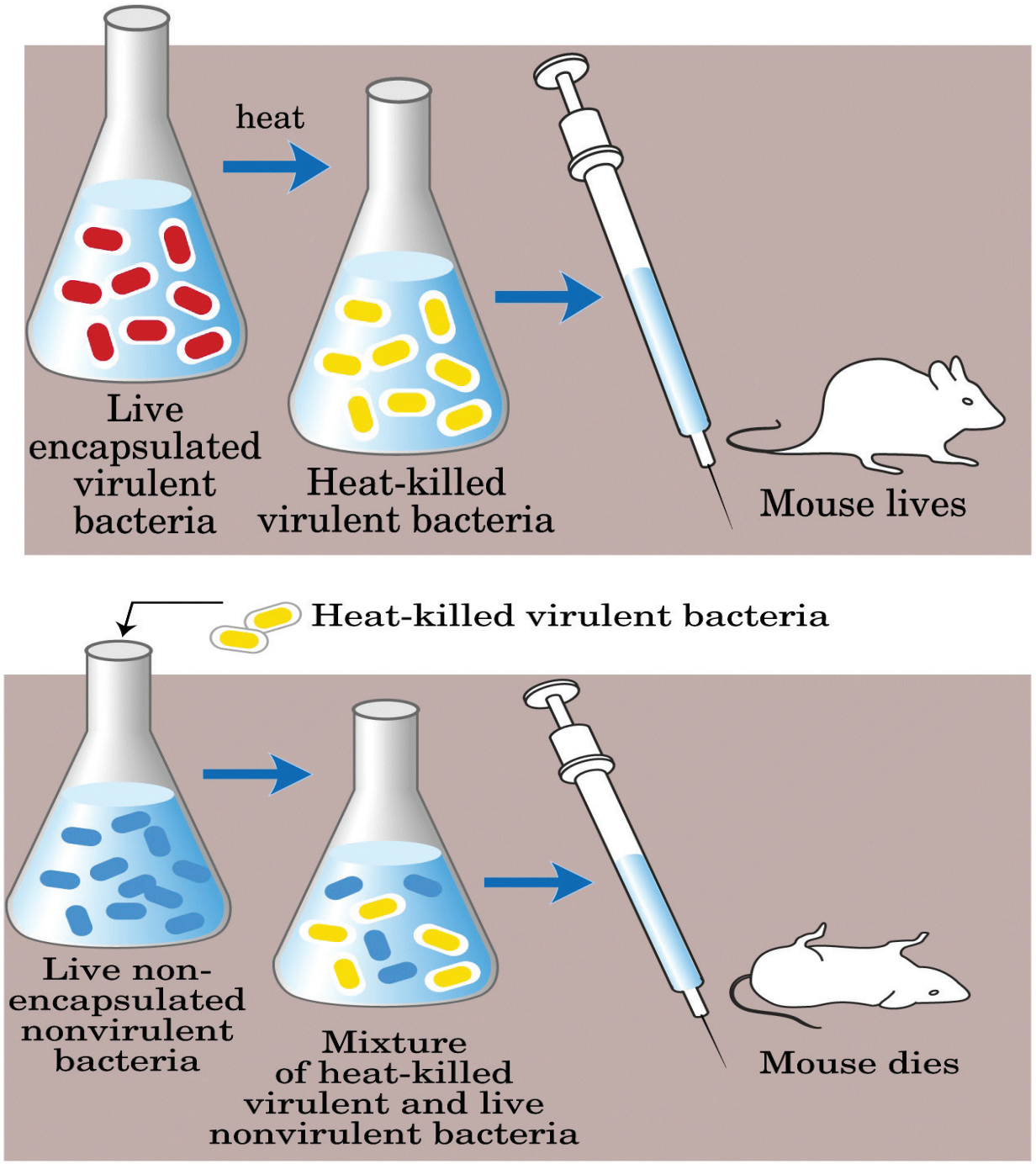
Experiments performed by **F. Griffith** in 1928 demonstrated that a “transforming factor” was able to confer virulence to avirulent strains of *Streptococcus pneumoniae.*



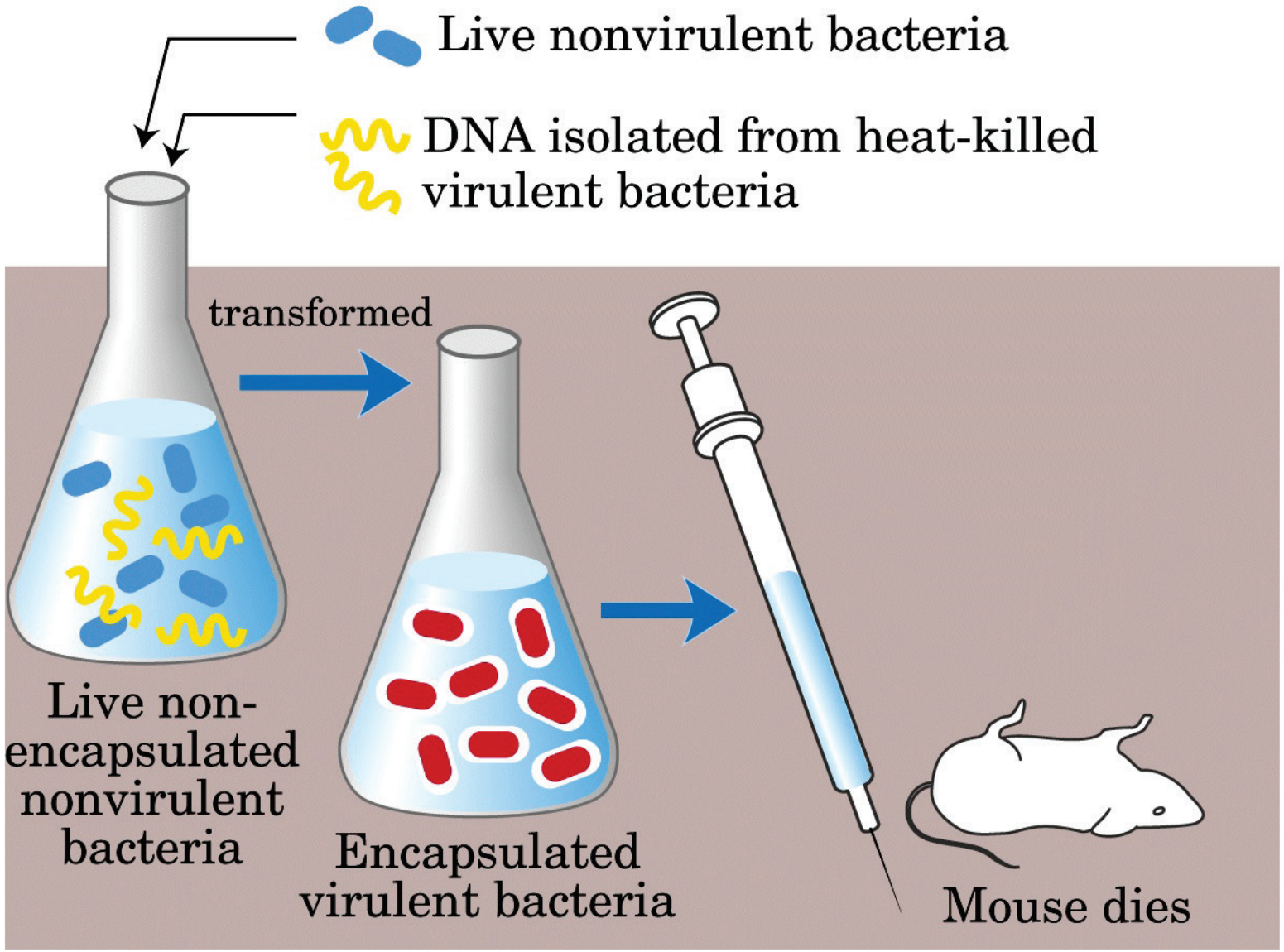
Frederick Griffith

(Hale, UK 1879-1941)



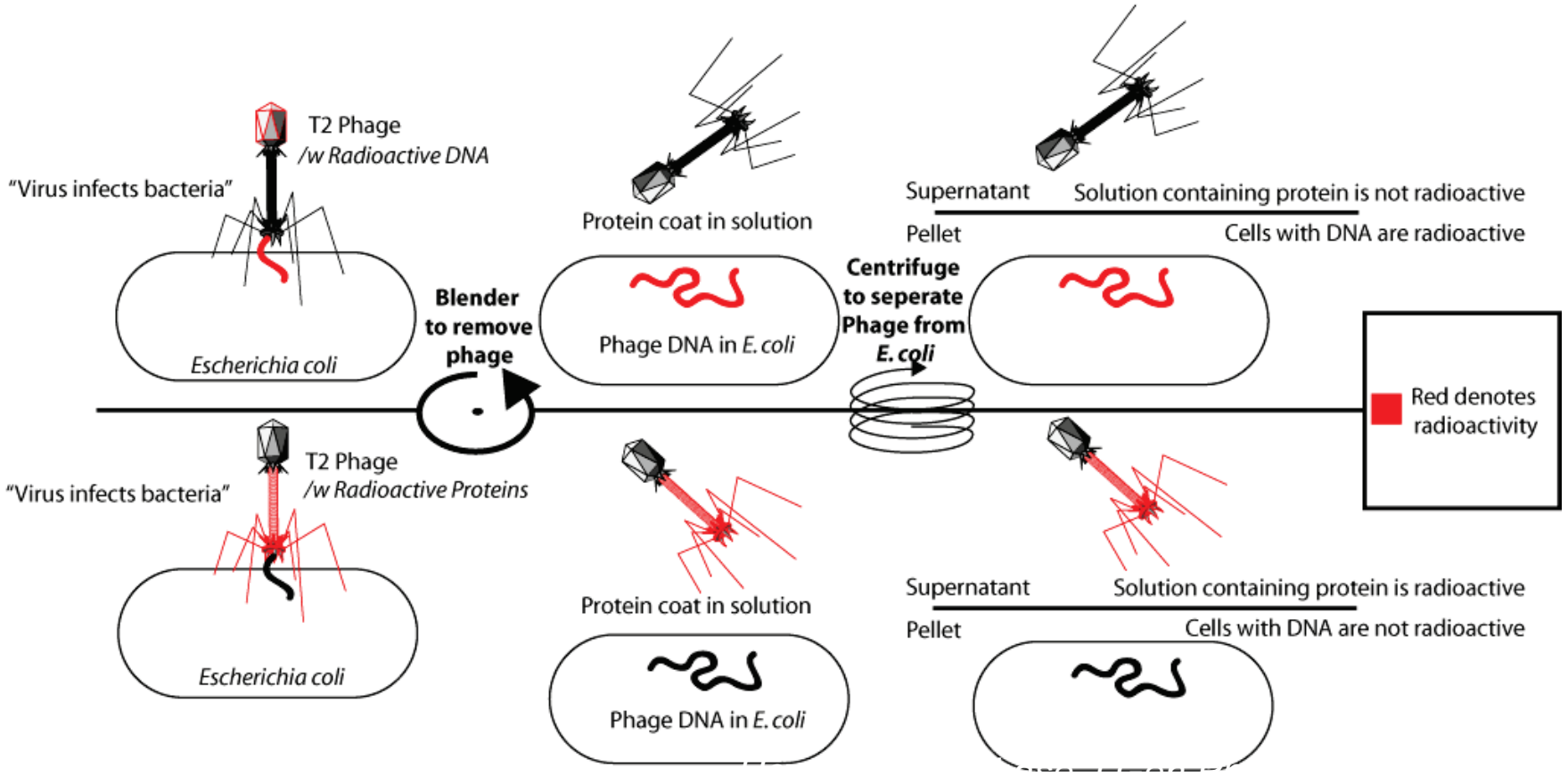

heat
Live encapsulated virulent bacteria
Heat-killed virulent bacteria
Mouse lives
Heat-killed virulent bacteria
Live non-encapsulated nonvirulent bacteria
Mixture of heat-killed virulent and live nonvirulent bacteria
Mouse dies

In 1944, Oswald Avery, Colin MacLeod and Maclyn McCarty identified this “transforming factor” as DNA.

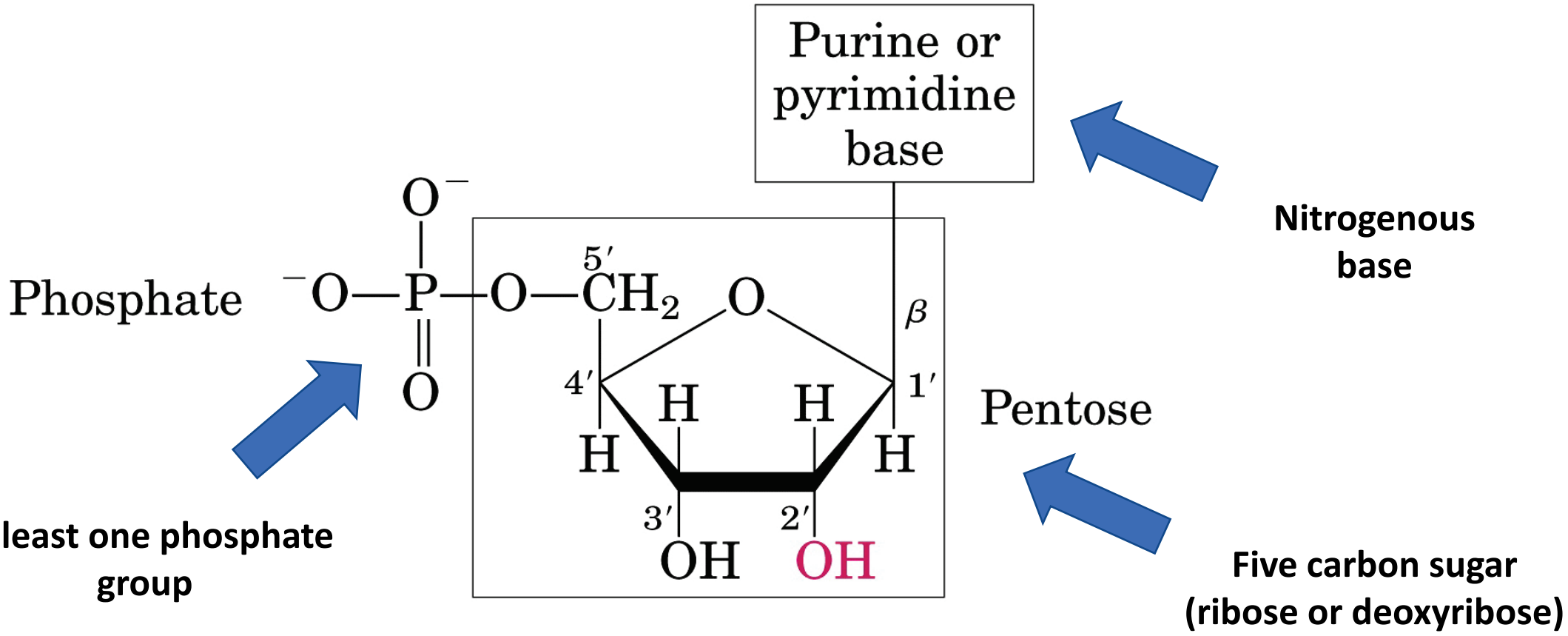


In 1952 **Alfred Hershey** and **Martha Chase** performed the definitive experiment demonstrating that DNA is the genetic material in living organisms.

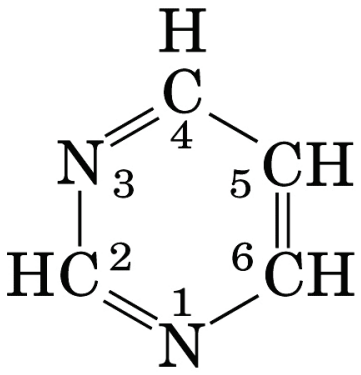
Proteins were labeled with ^{35}S and DNA was labeled with ^{32}P in T2 bacteriophage.



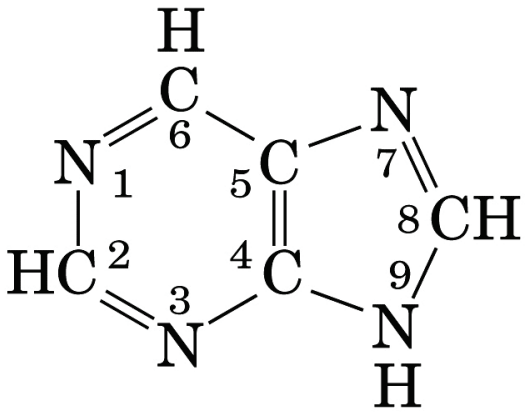
Nucleic acids are linear polymers formed by nucleotides



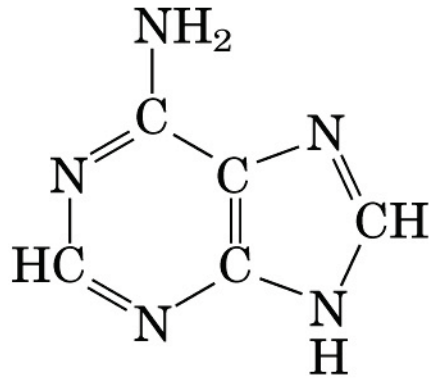
Nitrogenous bases (nuleobases)



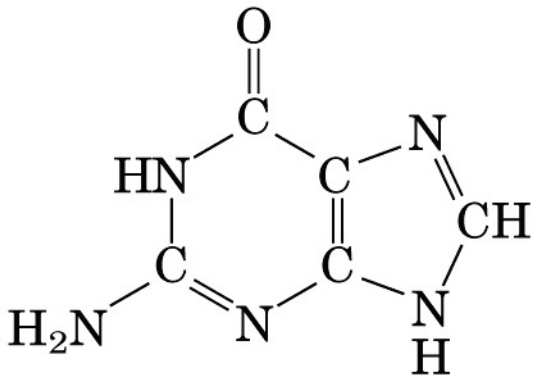
Pyrimidine



Purine

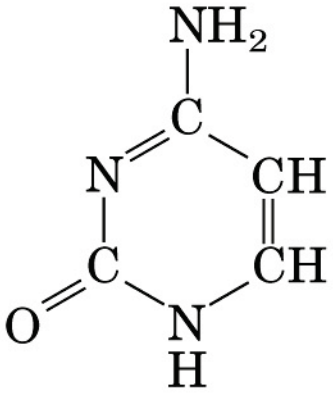


Adenine

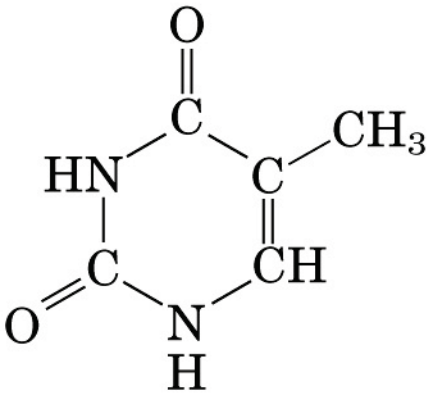


Guanine

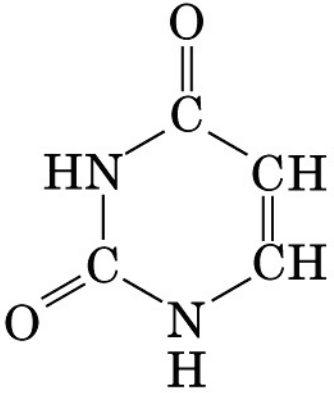
Purines



Cytosine



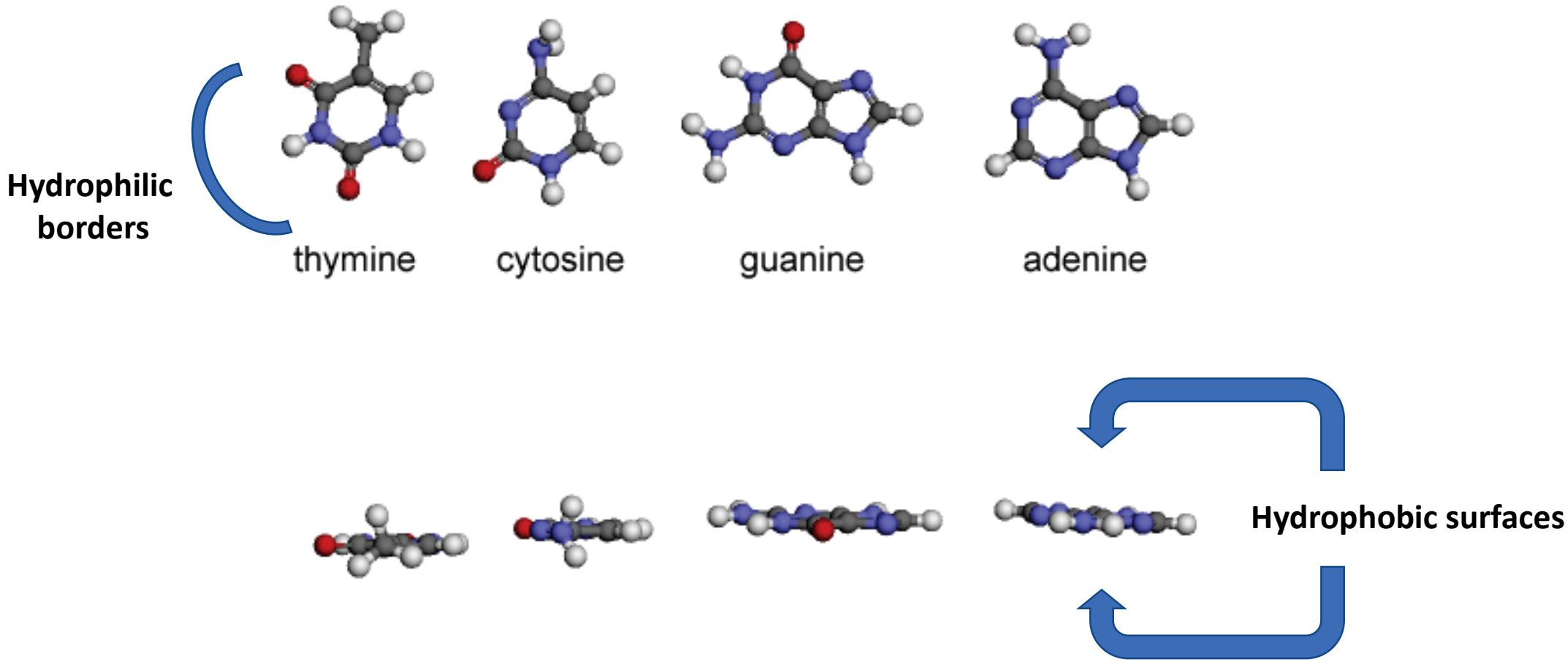
Thymine (DNA)



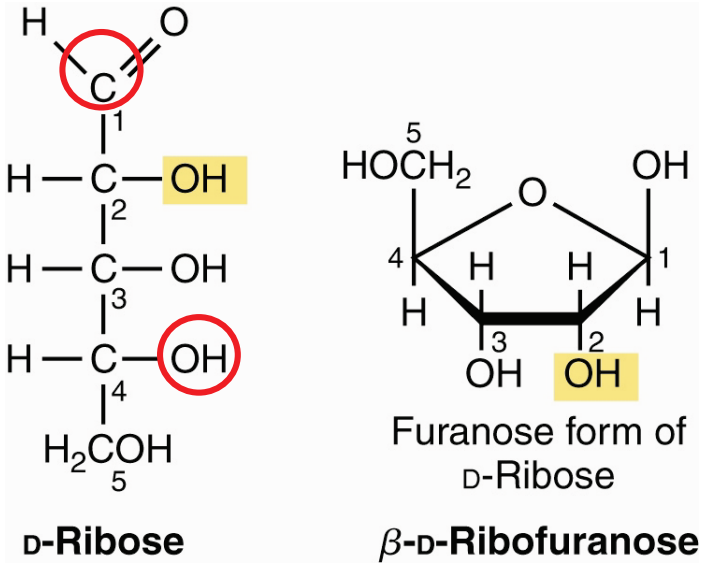
Uracil (RNA)

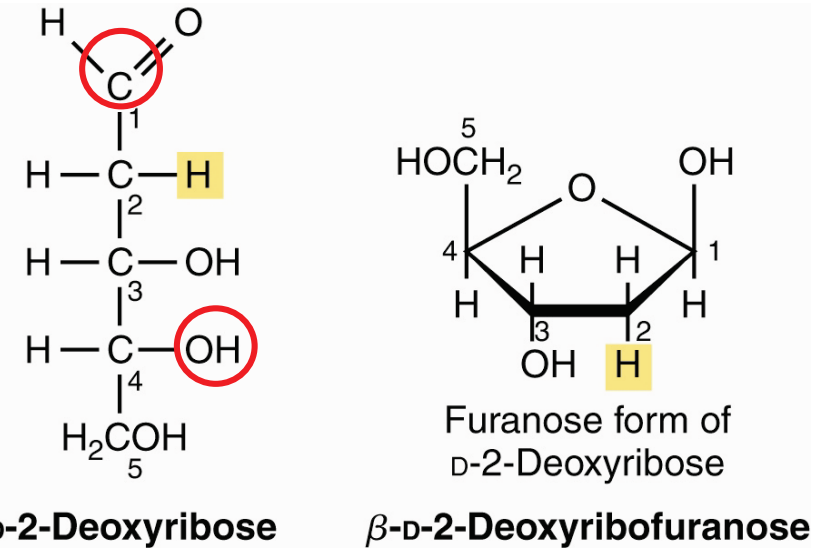
Pyrimidines

Nitrogenous bases

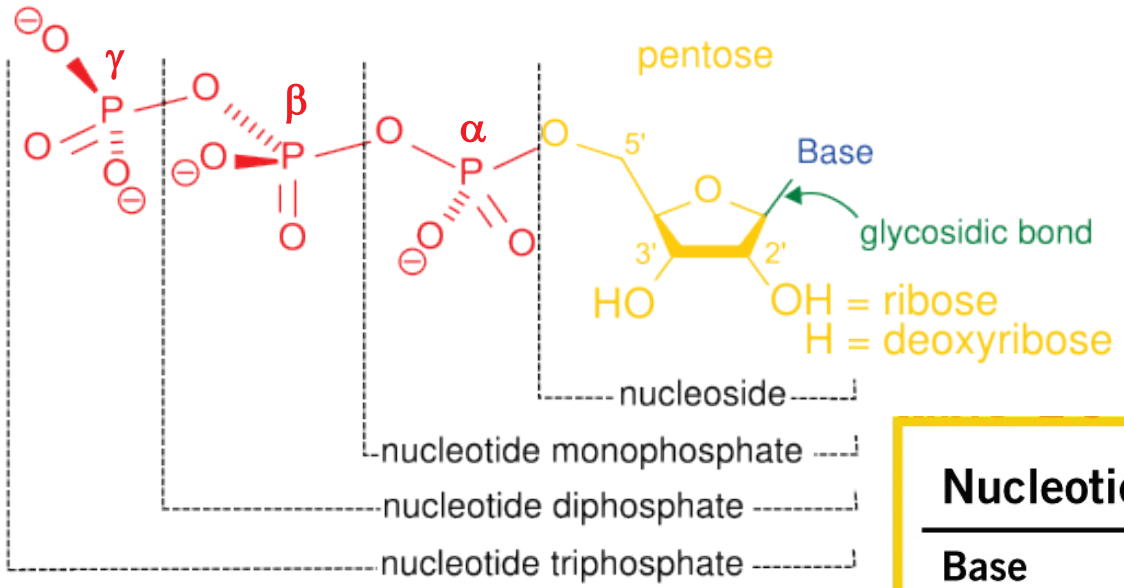


Five carbon sugars (pentoses) in nucleotides



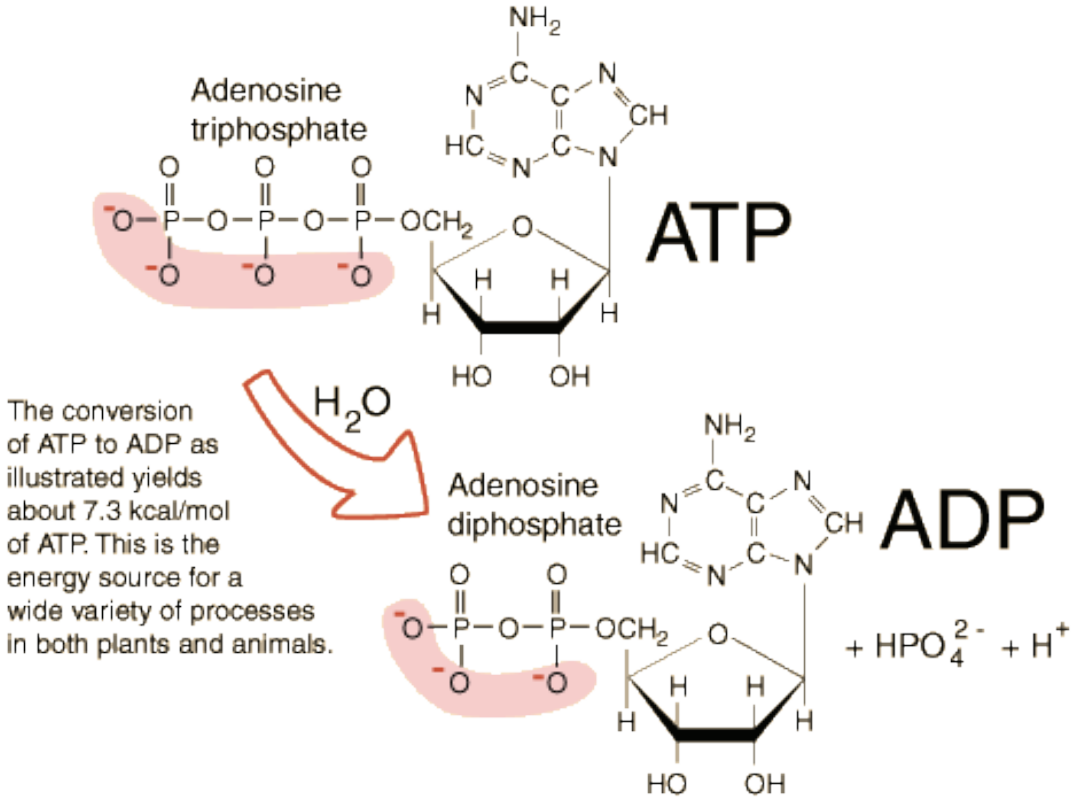


Formation of a hemiacetal bond

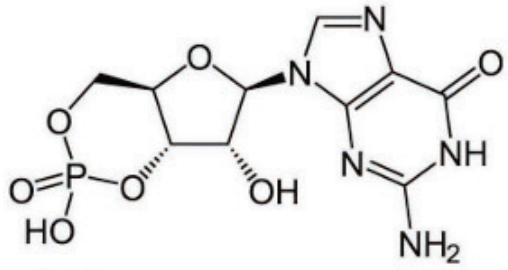


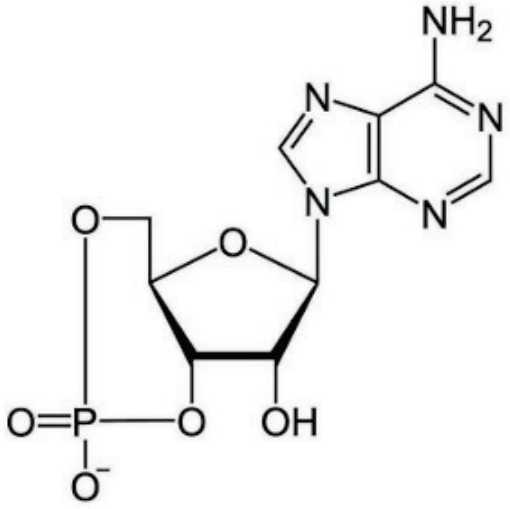
Nucleotide and Nucleic Acid Nomenclature

Base	Nucleoside*	Nucleotide*	Nucleic acid
Purines			
Adenine	Adenosine	Adenylate	RNA
	Deoxyadenosine	Deoxyadenylate	DNA
Guanine	Guanosine	Guanylate	RNA
	Deoxyguanosine	Deoxyguanylate	DNA
Pyrimidines			
Cytosine	Cytidine	Cytidylate	RNA
	Deoxycytidine	Deoxycytidylate	DNA
Thymine	Thymidine or deoxythymidine	Thymidylate or deoxythymidylate	DNA
Uracil	Uridine	Uridylate	RNA

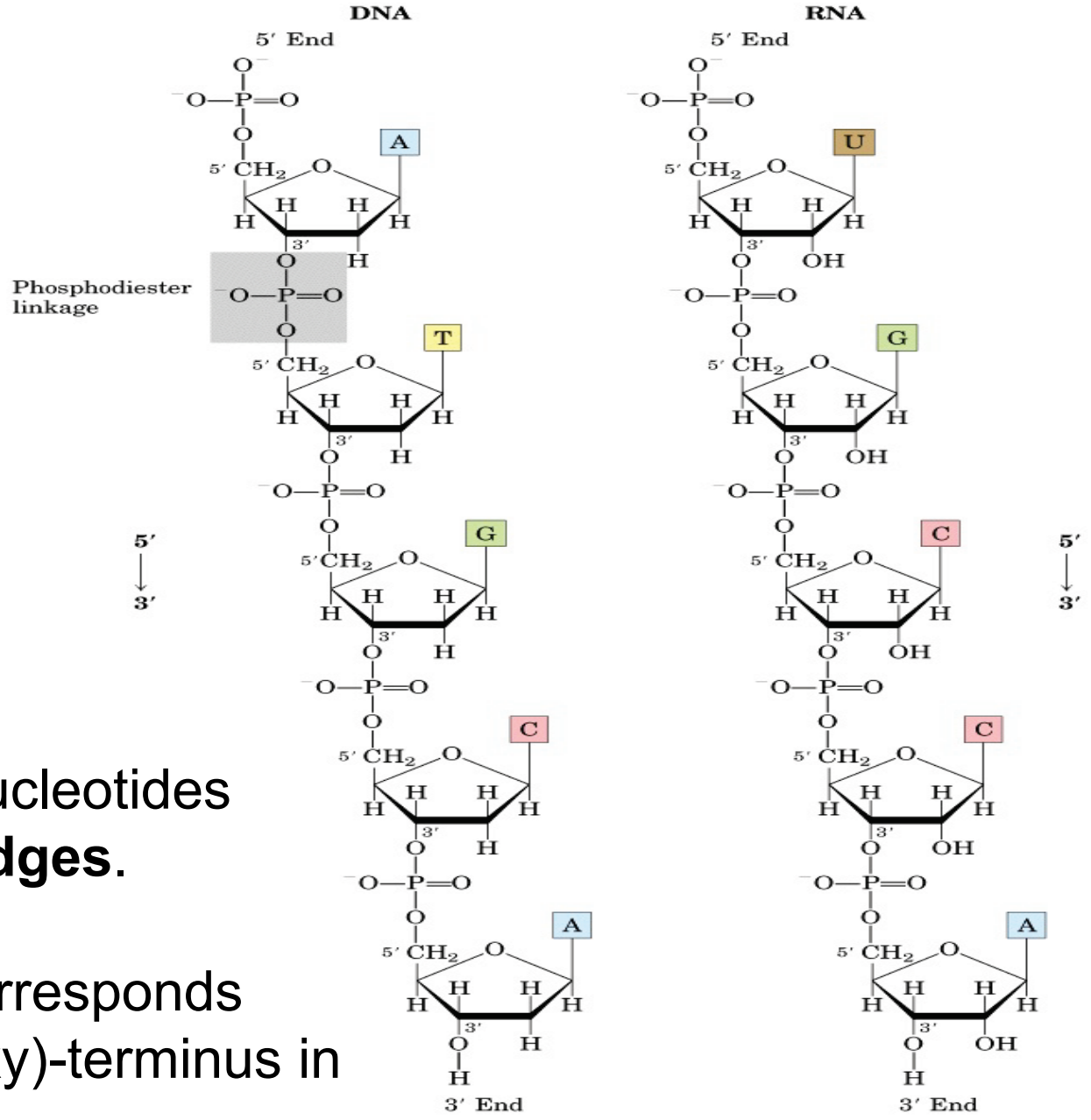


Diverse biological functions of free nucleotides





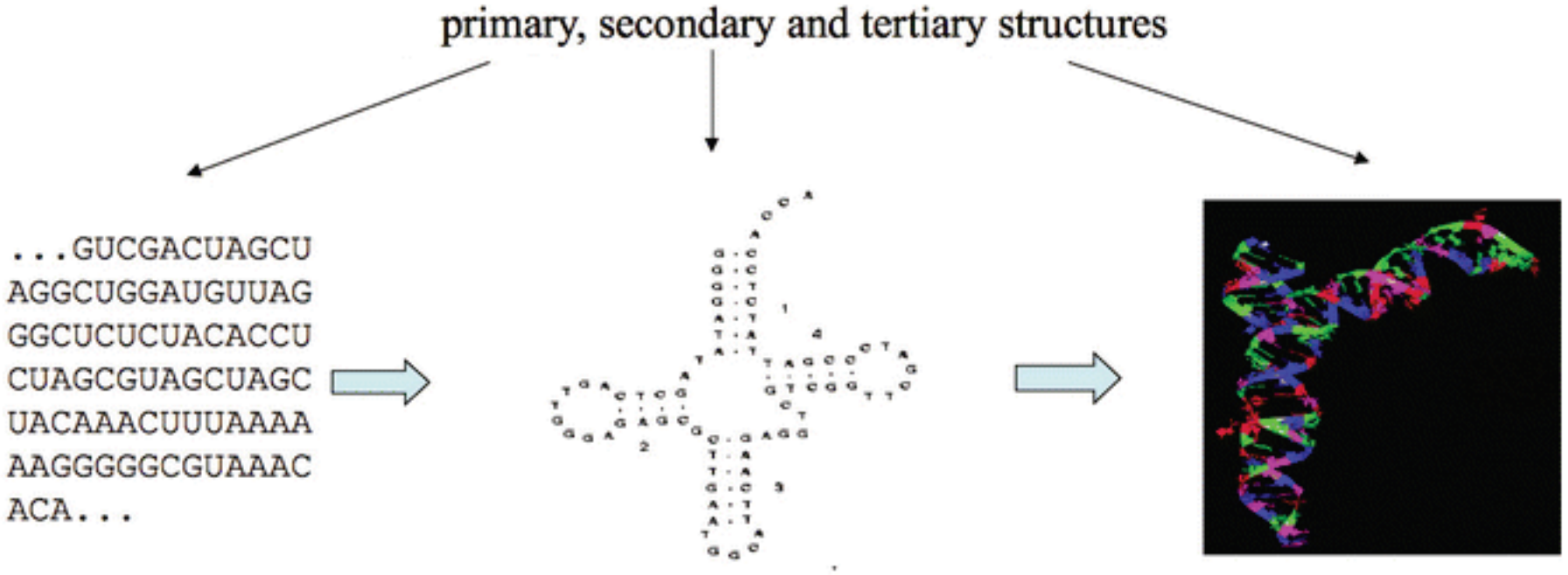
Polynucleotides



- Nucleic acids are linear polymers of nucleotides linked 3' to 5' by **phosphodiester bridges**.
- Sequence is <u>always</u> read 5' to 3'
- In terms of genetic information, this corresponds to the N (amino)-terminus to C (carboxy)-terminus in proteins.

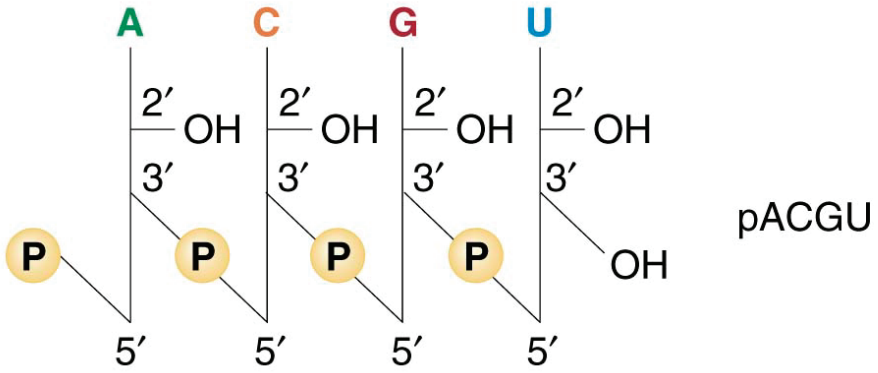
Level of structure in nucleic acids

- Primary structure: order of bases in polynucleotide
- Secondary structure: 3D conformation of the backbone
- Tertiary structure: supercoiling of the molecule



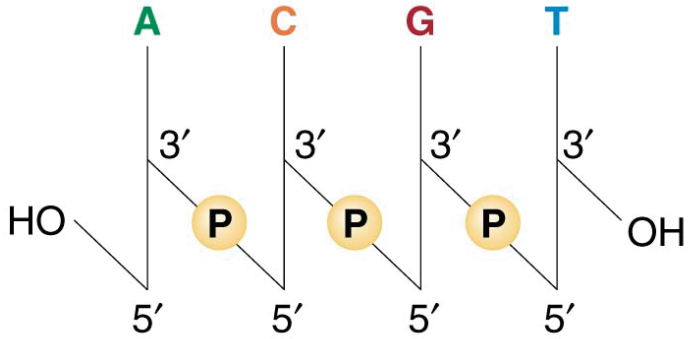
Primary structure

RNA



pACGU

DNA



5′ → 3′

d(ACGT)

Aligned sequences

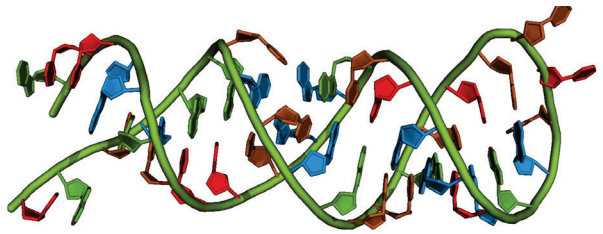
Human	ACA TTATGGACAGGTAAGTAAAAAACATATT
Chimpanzee	ACA TTATGGACAGGTAAGTAAAAAACATATT
Macaque	ATATACATTACGGACAGGTAAGTAAAAACAT

RNA secondary structure

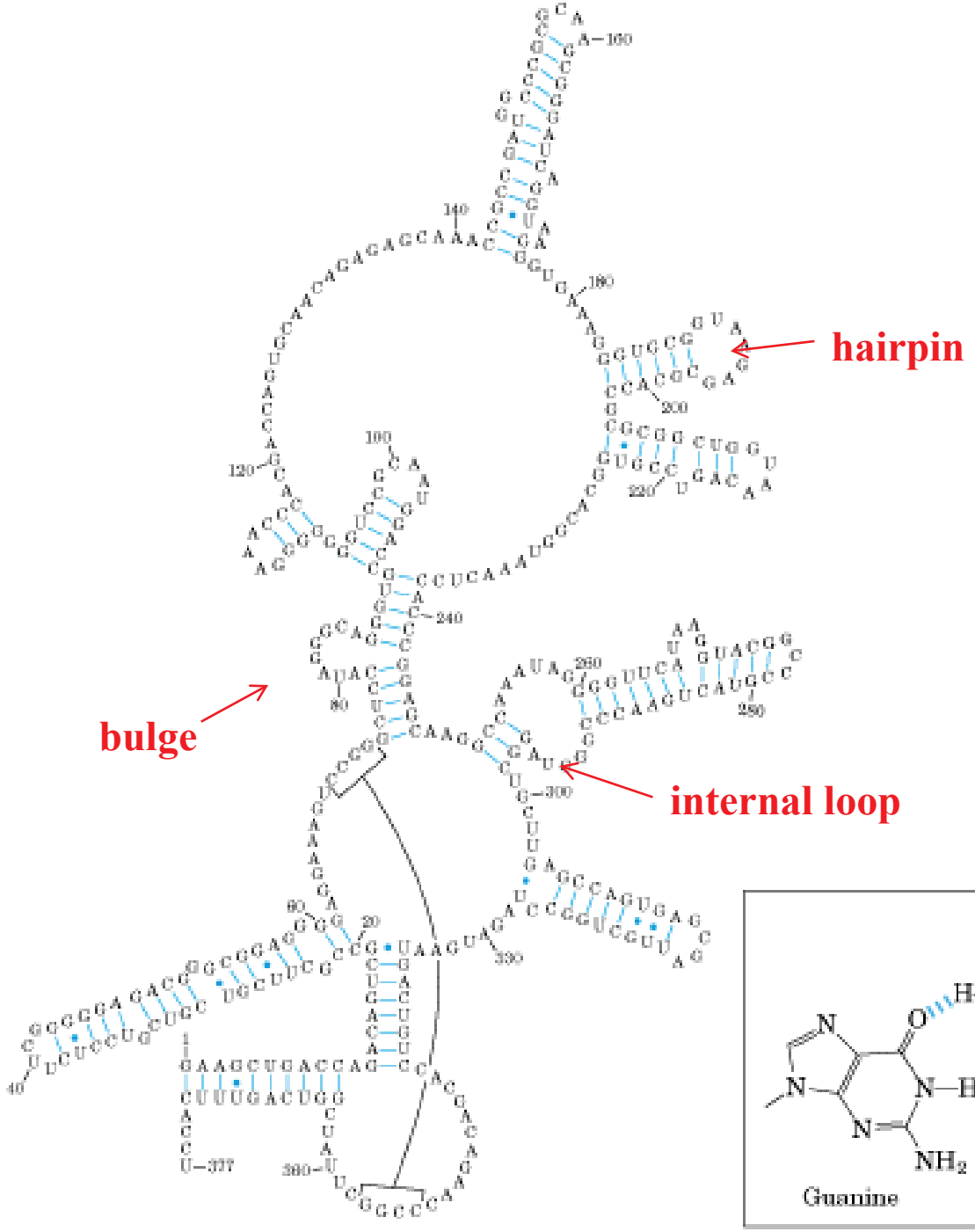
- RNA molecules are **single-stranded** but can adopt complex and unique secondary structures.

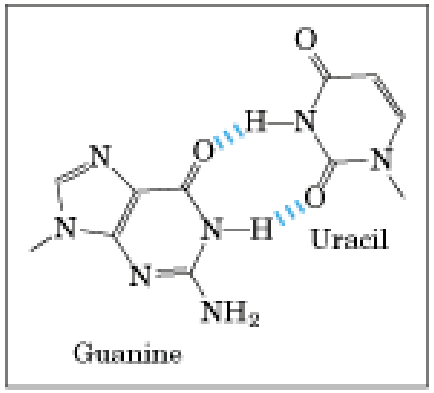
- RNA molecules can base pair with complementary DNA or RNA sequences.

- G pairs with C and U. A pairs with U.

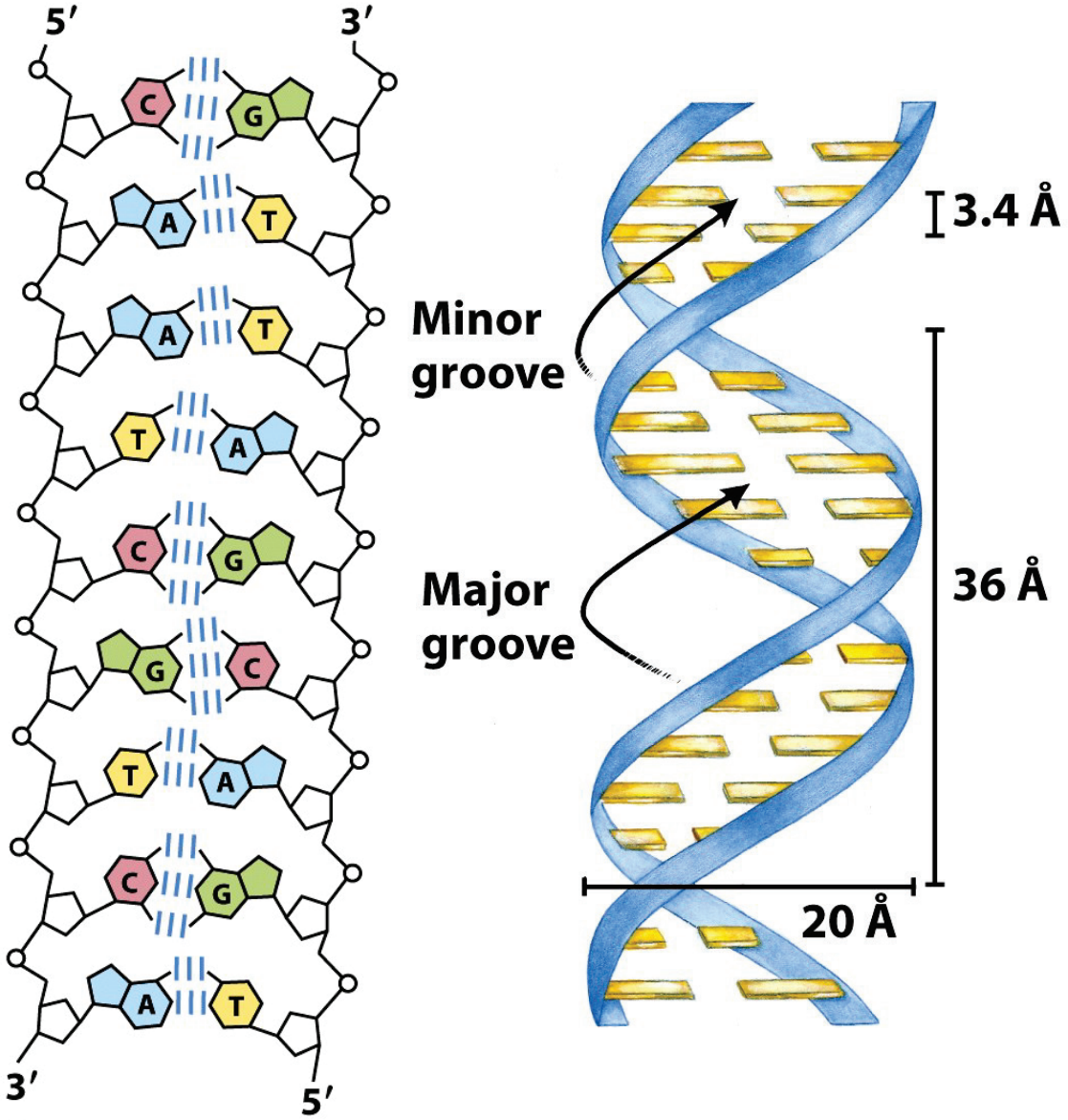


Hairpin double helix

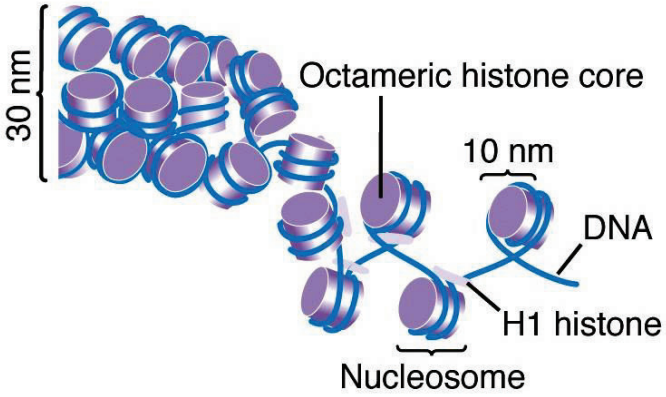




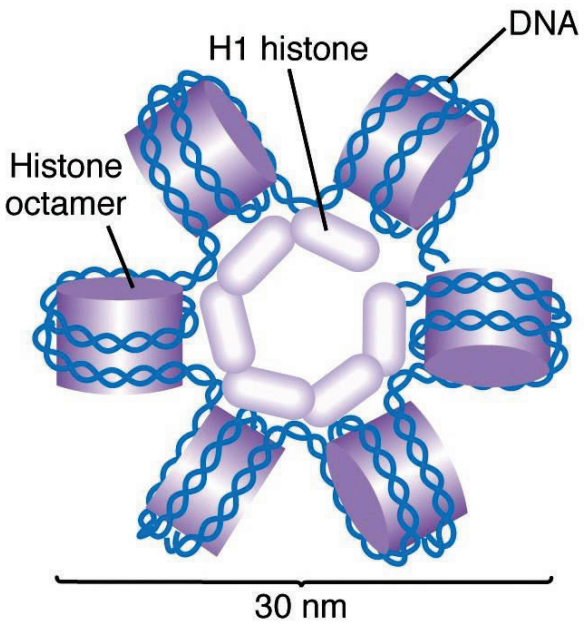
DNA secondary structure



DNA tertiary structure

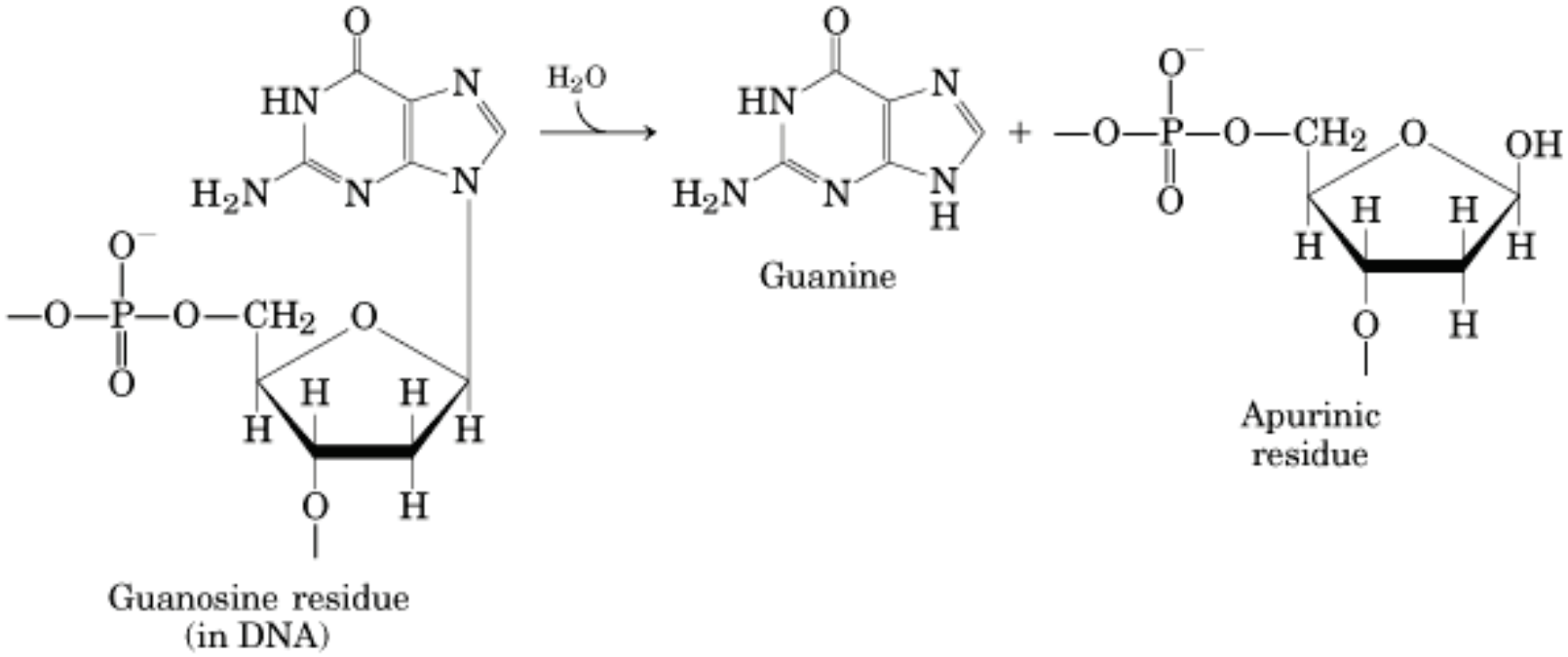


Supercoiling



Nucleic acids stability

Under physiological conditions the N-β-glycosidic bond, mainly in purines, can undergo **spontaneous hydrolysis** (1 every 10^5 purines/24h), generating the so-called “apurinic sites”. Experimentally this reaction is enhanced by acidic pH. There is a link between spontaneous mutation, aging, and carcinogenesis.



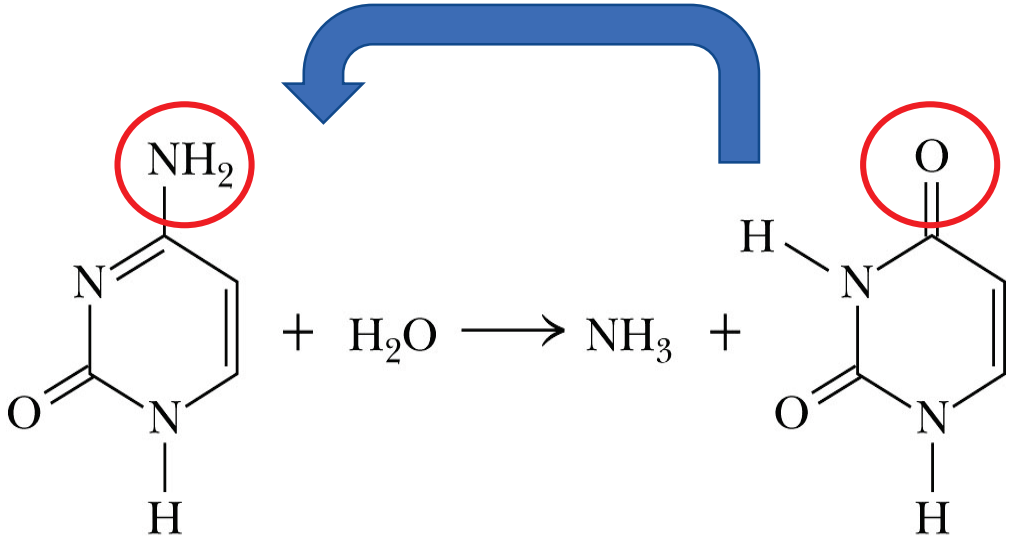
Nucleic acids stability

Under physiological conditions exocyclic amino groups can be lost in the so-called “deamination process”. **Cytosine deamination** results in the formation of **uracil** (1 every 10^7 citidine residues/24h). This would result in approximately 100 mutations per day in a cell. Deamination of adenine and guanine is 100-fold slower.

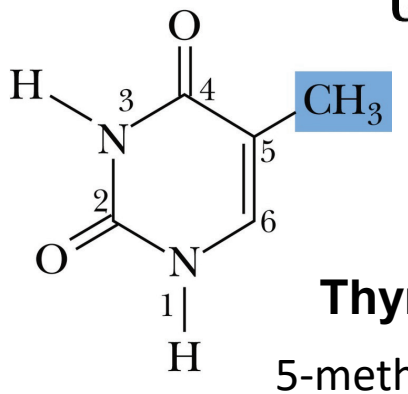
Uracils in DNA are recognized by repair mechanisms and are replaced by **cytosines**.

Why does DNA contain thymine and not uracil?

Repair mechanisms



Cytosine **Uracil**



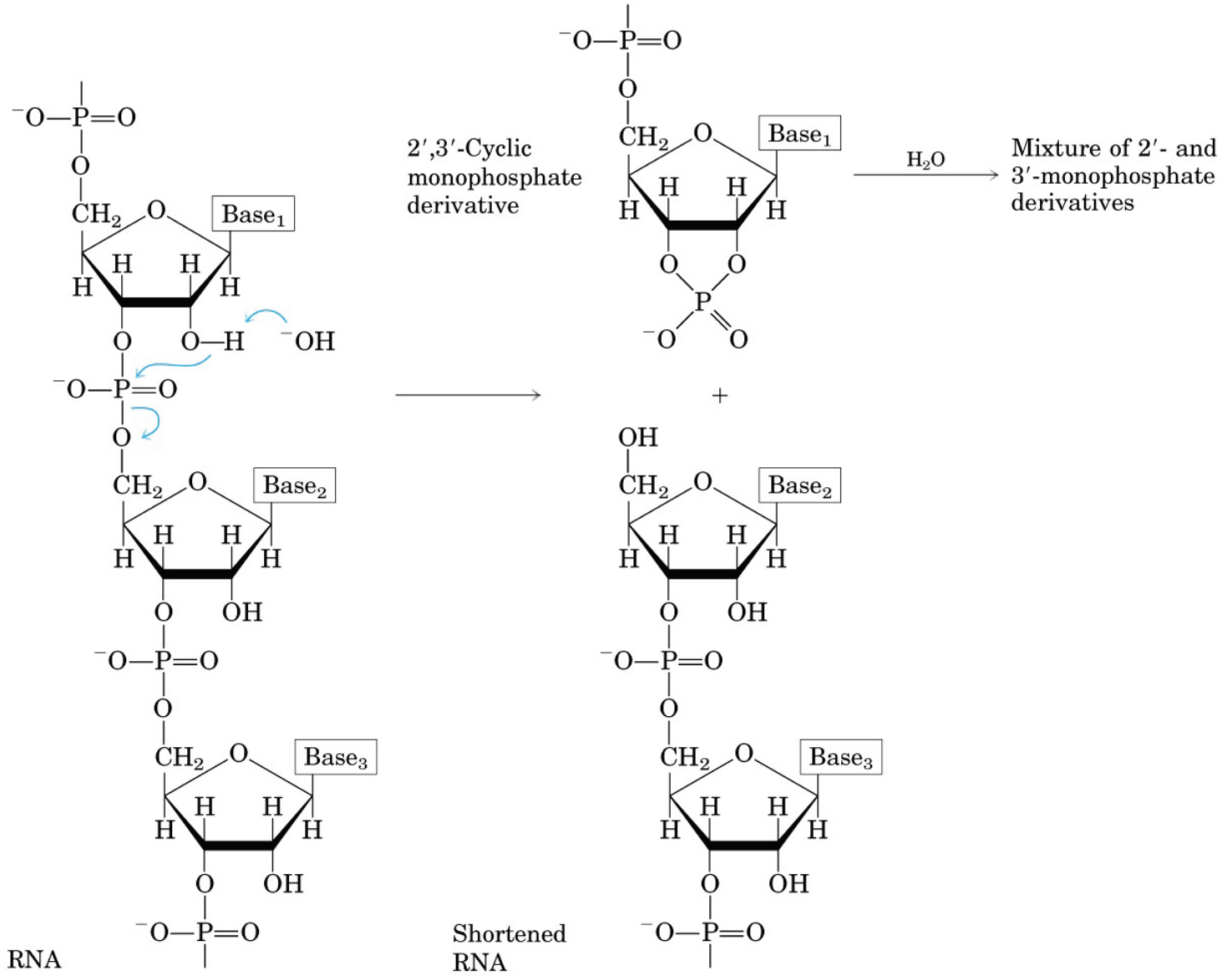
Thymine

5-methyl uracil

If DNA contained uracil, during replication of DNA the uracils would be base-paired with adenine.
Deaminated cytosines (”spontaneously formed uracils”) would also be base-paired with adenine. This would decrease the number of G-C base pairs over time and increase the number of A-U base pairs.
Eventually all the G-C base pairs could be lost, and the genetic code would not exist as we know it.
Repair enzymes would not be able to recognize natural uracils from “mutant” uracil.
Nature solved this dilema by replacing uracil by 5-methyl uracil, i.e. thymine.

Nucleic acids stability

DNA is very stable in aqueous solutions at neutral pH. The high stability of phosphodiester bonds makes **DNA** the ideal molecule for the storage of genetic information. **RNA** is much more prone to the hydrolysis of phosphodiester bonds



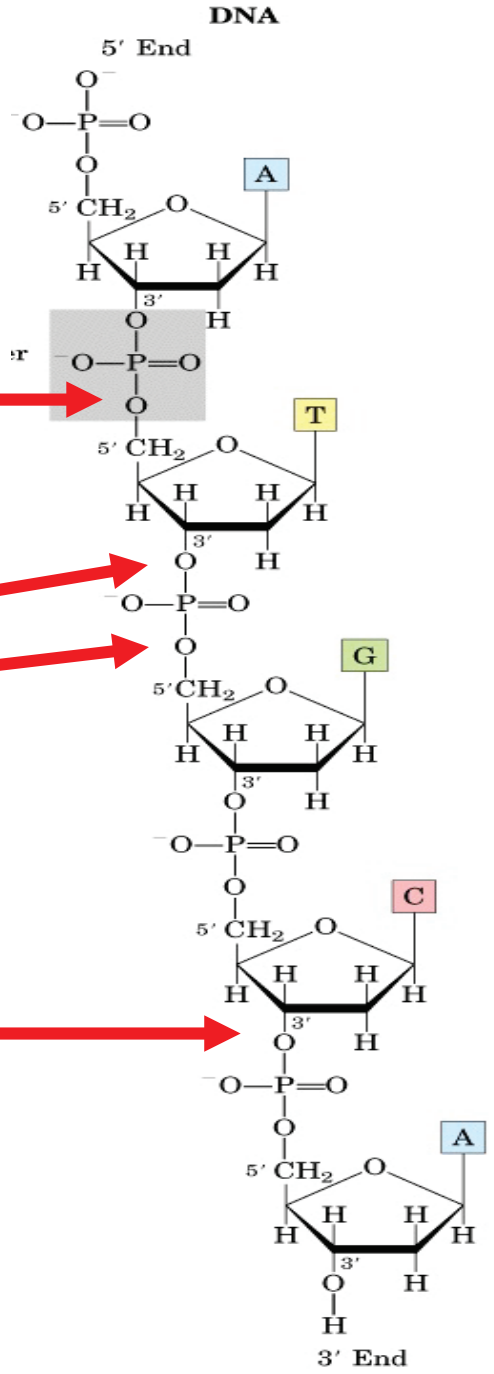
NUCLEASES:

Enzymes with phosphodiesterase activity that digest nucleic acids.

EXONUCLEASES (5')

ENDONUCLEASES
DNase I, DNase II

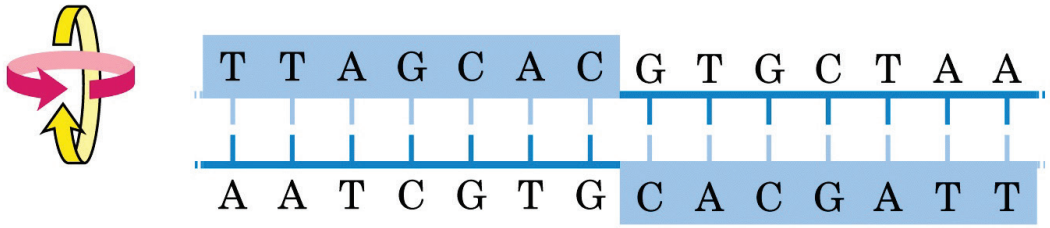
EXONUCLEASES (3')

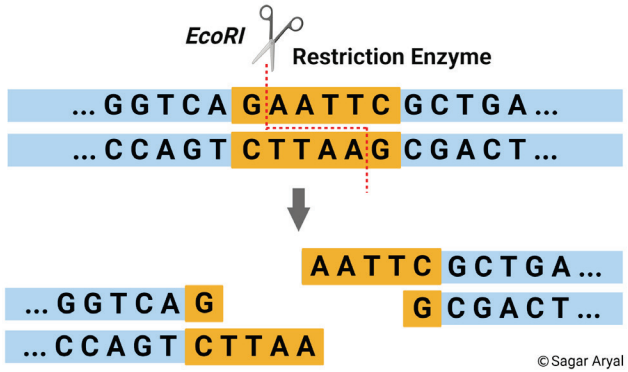


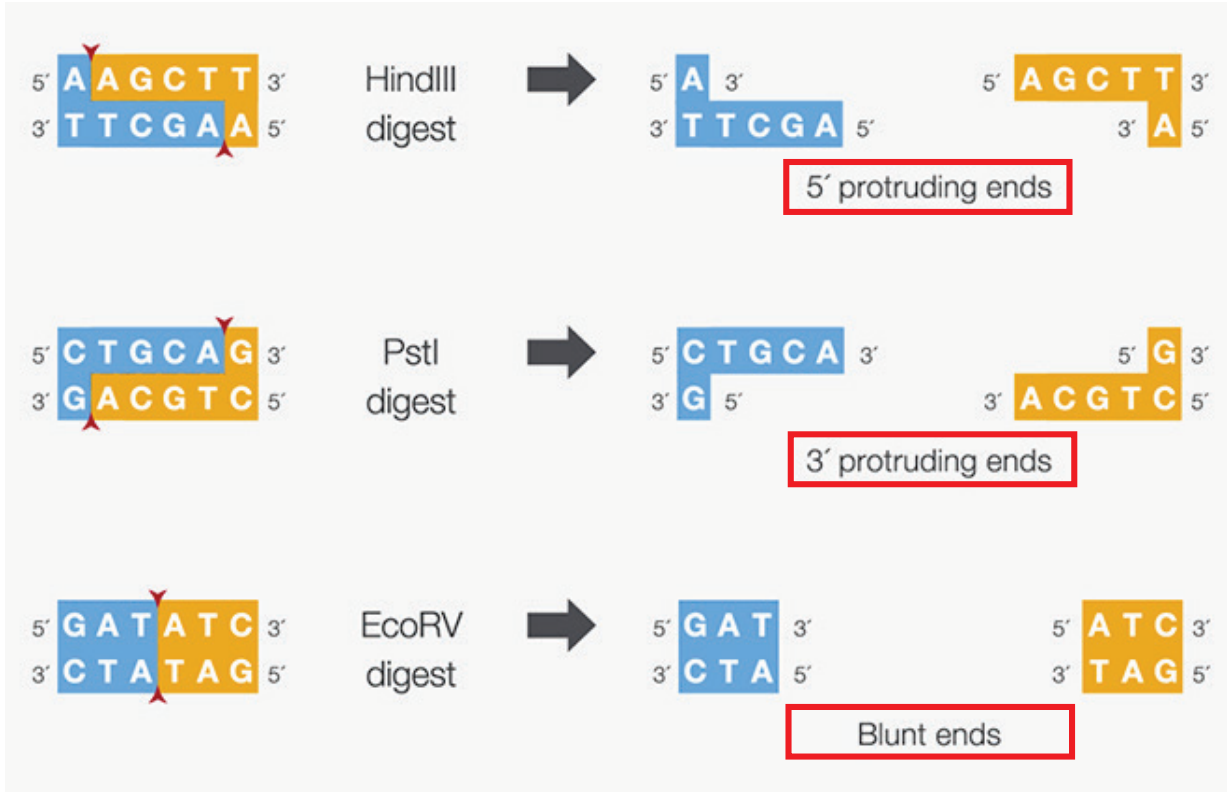
RESTRICTION ENDONUCLEASES

- Bacteria have learned to "restrict" the possibility of attack from foreign DNA by means of "restriction enzymes".
- Enzymes may recognize 4, 6 or more bases in selecting sites for cleavage.
- No ATP requirement.
- Recognition sites in dsDNA have a 2-fold axis of symmetry (palindromic sequences).
- Cleavage can leave staggered/protruding ("sticky") ends or can produce "blunt" ends.

Palindrome



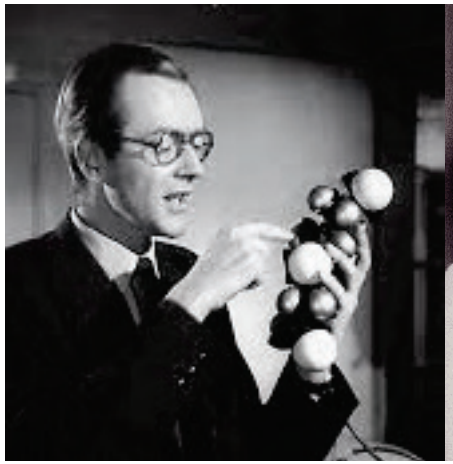


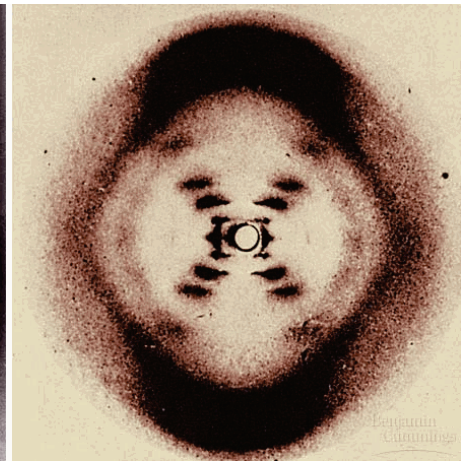


THE STRUCTURE OF DNA

Maurice Wilkins
(New Zealand 1916- UK 2004)

Rosalind Franklin
(UK 1920-1958)





Studies carried out during the first half of the XXth century on DNA X **rays diffraction pattern** and **chemical composition** led to propose a secondary structure for DNA.

The three fundamental evidences that allowed the establishment of the secondary structure of DNA were:

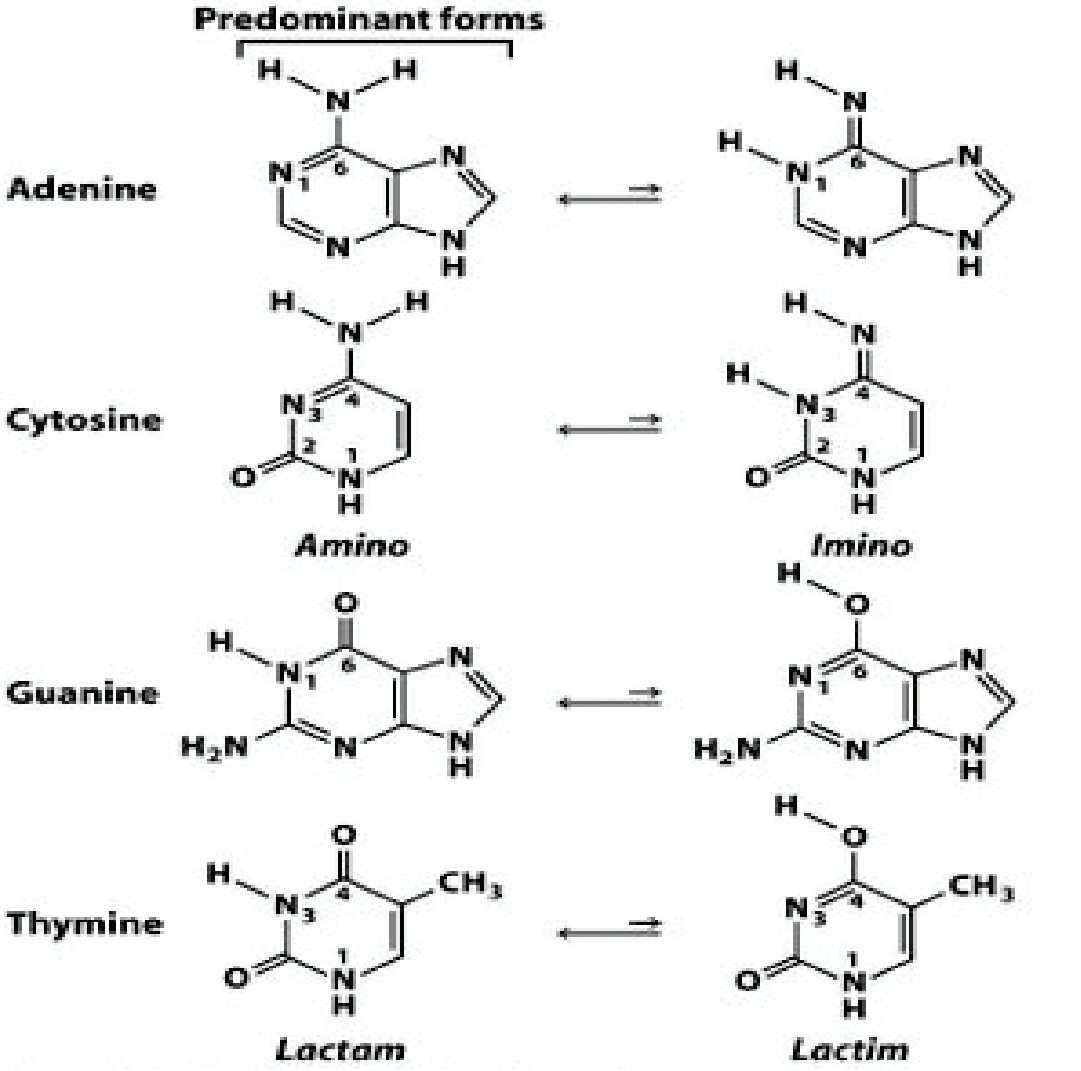
1. X rays diffraction data suggested that DNA had a double helical linear structure, and that both strands were oriented in an antiparallel fashion. DNA molecules exhibit **two periodicities** repeating along the length of the fiber, a primary repeat of 3.4 Å and a secondary repeat of 34 Å.

2. Studies on the **chemical composition** of DNA informed about the relative abundance of each nucleobase.

3. Nucleobases are preferentially found in their keto and amino **tautomeric forms**, with preference over the enol and imino **forms**. This finding allowed the establishment of the correct base pairs by means of hydrogen bonds.

TAUTOMERIC FORMS OF NUCLEOBASES

Free pyrimidine and purine bases may exist in two or more **tautomeric forms** depending on the pH. Certain tautomeric forms predominate at neutral pH, and these are the structures commonly depicted for the four purines and pyrimidines. These are the tautomers that are present in the bases in DNA and RNA.

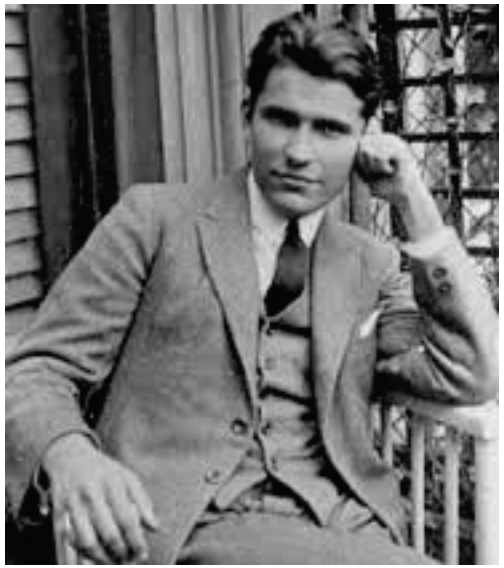


AMINO and KETO forms

IMINO and ENOL forms

Erwin Chargaff

(Austria 1905-USA 2002)



Erwin Chargaff findings on the **chemistry of DNA** (1950-52):

1. The nucleobase composition of DNA generally varies from one species to another.

2. DNA specimens isolated from different tissues of the same species have the same base composition.

3. The base composition of DNA in a given species does not change with age, nutritional stage, or changing environment.

4. In all cellular DNAs, regardless of the species, the number of adenosine residues is equal to the number of thymidine residues (**A=T**), and the number of guanosine residues is equal to the number of cytidine residues (**G=C**). Therefore the sum of purine residues equals the sum of the pyrimidine residues: **A+G = T+C**.

All this structural (X rays diffraction patterns) and chemical information on DNA was put together by J.D. Watson and F. Crick in their 1953 proposal for the structure of this molecule.

equipment, and to Dr. G. E. R. Deacon and the captain and officers of R.R.S. *Discovery II* for their part in making the observations.

[1] Young, F. B., Gerrard, H., and Jevons, W., *Phil. Mag.*, **40**, 149 (1920).
[2] Longuet-Higgins, M. S., *Mon. Not. Roy. Astro. Soc., Geophys. Supp.*, **5**, 285 (1949).
[3] Von Arx, W. S., Woods Hole Papers in Phys. Oceanog. Meteor., **11** (3) (1950).
[4] Ekman, V. W., *Arkiv. Mat. Astron. Fysik. (Stockholm)*, **2** (11) (1905).

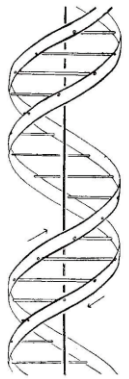
MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey[1]. They kindly made their manuscript available to us in advance of publication. Their model consists of three intertwined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion, this structure is unsatisfactory for two reasons: (1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.



This figure is purely diagrammatic. The two ribbons symbolize the two phosphate—sugar chains, and the horizontal rods the pairs of bases holding the chains together. The vertical line marks the fibre axis

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram). We have made the usual chemical assumptions, namely, that each chain consists of phosphate diester groups joining β-D-deoxyribofuranose residues with 3′,5′ linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow right-handed helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions. Each chain loosely resembles Furberg's[2] model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furberg's 'standard configuration', the sugar being roughly perpendicular to the attached base. There is a residue on each chain every 3·4 A. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 A. The distance of a phosphorus atom from the fibre axis is 10 A. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical z-co-ordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations) it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain is given, then the sequence on the other chain is automatically determined.

It has been found experimentally[3,4] that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine, are always very close to unity for deoxyribose nucleic acid.

It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

The previously published X-ray data[5,6] on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in the following communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereochemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

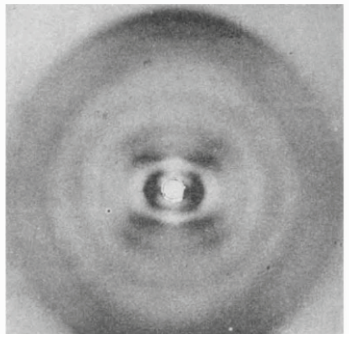
Full details of the structure, including the conditions assumed in building it, together with a set of co-ordinates for the atoms, will be published elsewhere.

We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. M. H. F. Wilkins, Dr. R. E. Franklin and their co-workers at King's College, London. One of us (J. D. W.) has been aided by a fellowship from the National Foundation for Infantile Paralysis.

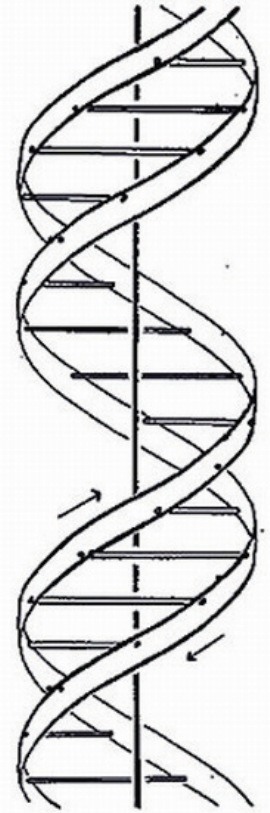
J. D. Watson
F. H. C. Crick

Medical Research Council Unit for the Study of the Molecular Structure of Biological Systems,
Cavendish Laboratory, Cambridge.
April 2.

[1] Pauling, L., and Corey, R. B., *Nature*, **171**, 346 (1953); *Proc. U.S. Nat. Acad. Sci.*, **39**, 84 (1953).
[2] Furberg, S., *Acta Chem. Scand.*, **6**, 634 (1952).
[3] Chargaff, E., for references see Zamenhof, S., Brawerman, G., and Chargaff, E., *Biochim. et Biophys. Acta*, **9**, 402 (1952).
[4] Wyatt, G. R., *J. Gen. Physiol.*, **36**, 201 (1952).
[5] Astbury, W. T., Symp. Soc. Exp. Biol. 1, Nucleic Acid, 66 (Camb. Univ. Press, 1947).
[6] Wilkins, M. H. F., and Randall, J. T., *Biochim. et Biophys. Acta*, **10**, 192 (1953).







This figure is purely diagrammatic. The two ribbons symbolize the two phosphate—sugar chains, and the horizontal rods the pairs of bases holding the chains together. The vertical line marks the fibre axis

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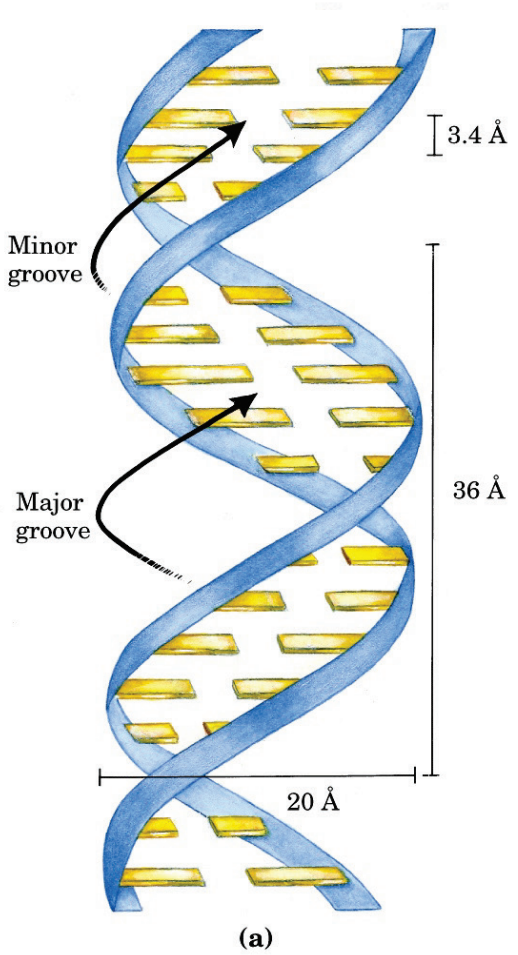
Secondary structure of DNA

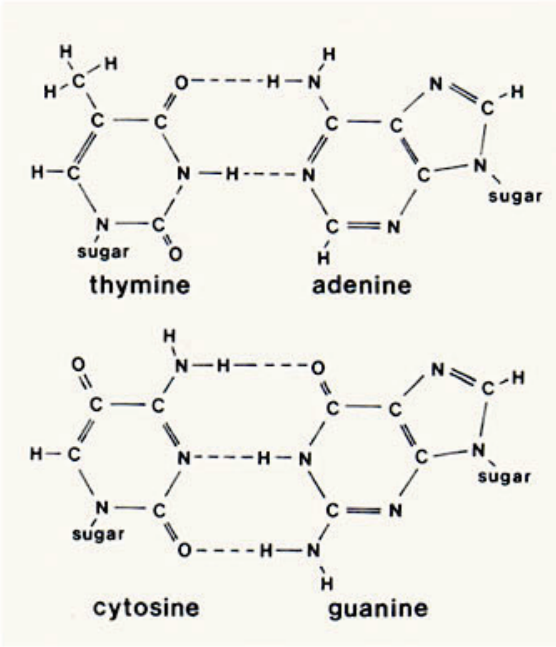
- **Base stacking** accounts for the 3.4 Å repeat along the length of the helix.

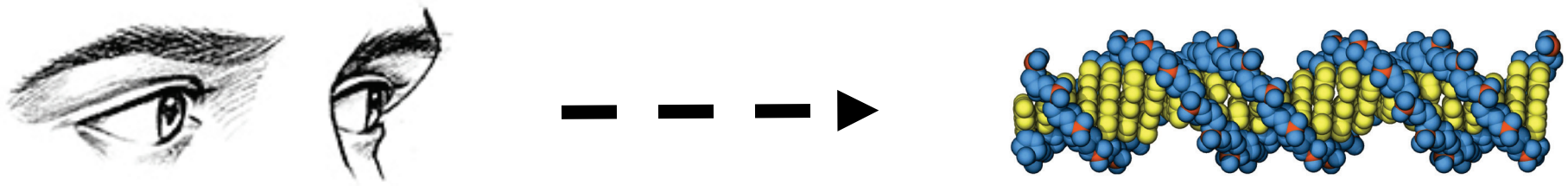
- The secondary repeat of about 34 Å was accounted for by **the presence of 10 base pairs in each complete turn of the double helix**. This was later modified to **10.5 base pairs** per turn for DNA in aqueous solution.

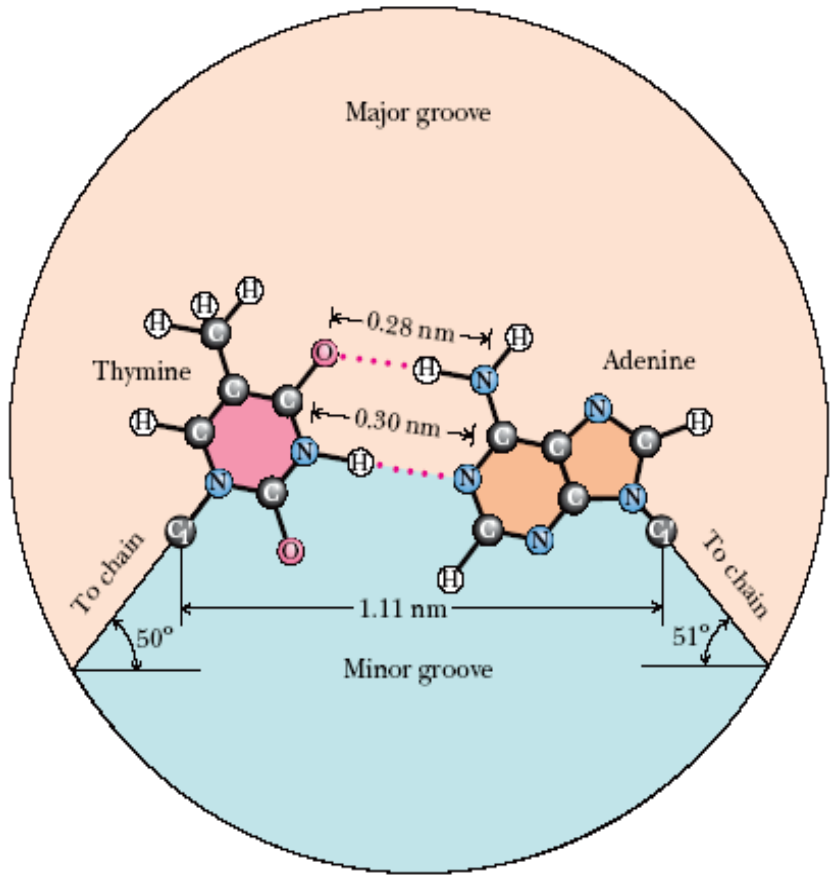
- In the Watson-Crick model, A/T and G/C **base pairing** was proposed based on the fact that these combinations of bases fit well inside the double helix.

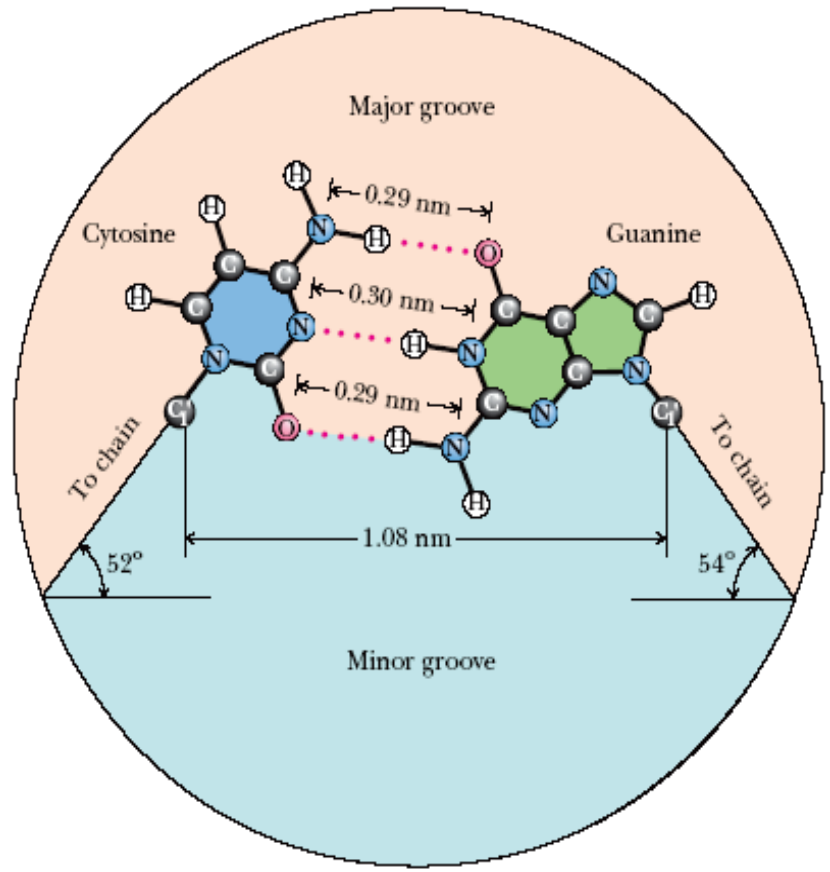
- Finally, the offset pairing of the two strands creates a **major and a minor groove** on the surface of the duplex.








Major groove
Thymine
Adenine
0.28 nm
0.30 nm
To chain
To chain
1.11 nm
50°
51°
Minor groove


Major groove
Cytosine
Guanine
0.29 nm
0.30 nm
0.29 nm
To chain
To chain
1.08 nm
52°
54°
Minor groove

Structural variation in DNA

- Double-helical **DNA is a very flexible molecule**. Considerable rotation is possible around several types of bonds in the phosphodeoxyribose backbone, and thermal fluctuation can produce bending, stretching, and unpairing (melting) of the strands.

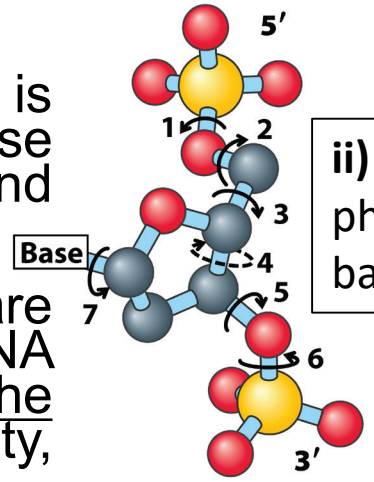
- Many significant **deviations from the Watson-Crick DNA structure** are found in cellular DNA, some or all of which may be important in DNA metabolism. Note that these structural variations generally do not affect the fundamental properties of DNA, such as strand complementarity, antiparallel strands, and the requirement for A/T and G/C base pairs.

- Structural variations in DNA are due to three things:

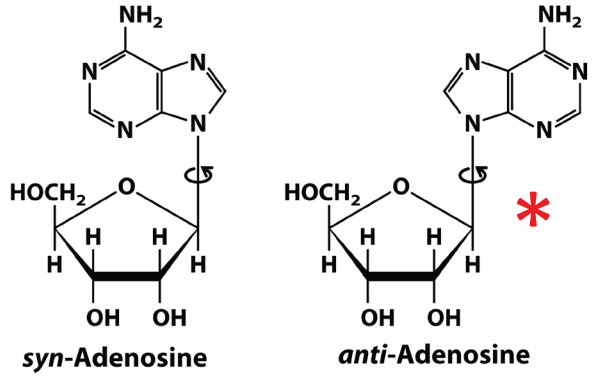
i) The different possible conformations of the deoxyribose,

ii) Rotation about the contiguous bonds that make up the phosphodeoxyribose backbone.

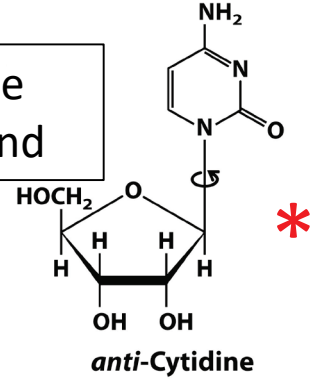
iii) Free rotation about the C-1'-*N*-glycosyl bond. Because of steric constraints, purines in purine nucleotides are restricted to two stable conformations with respect to deoxyribose, called syn and anti. Anti conformation is dominant in purines. Pyrimidines generally are restricted to the anti conformation because of steric interference between the sugar and the carbonyl oxygen at C-2 of the pyrimidine.



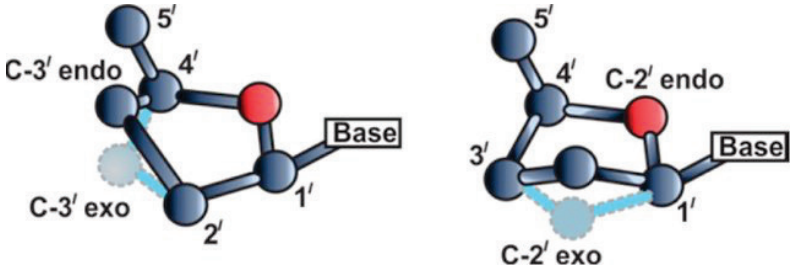
ii) Rotation of phosphodeoxyribose backbone bonds



iii) Rotation of the C-1'-N glycosyl bond



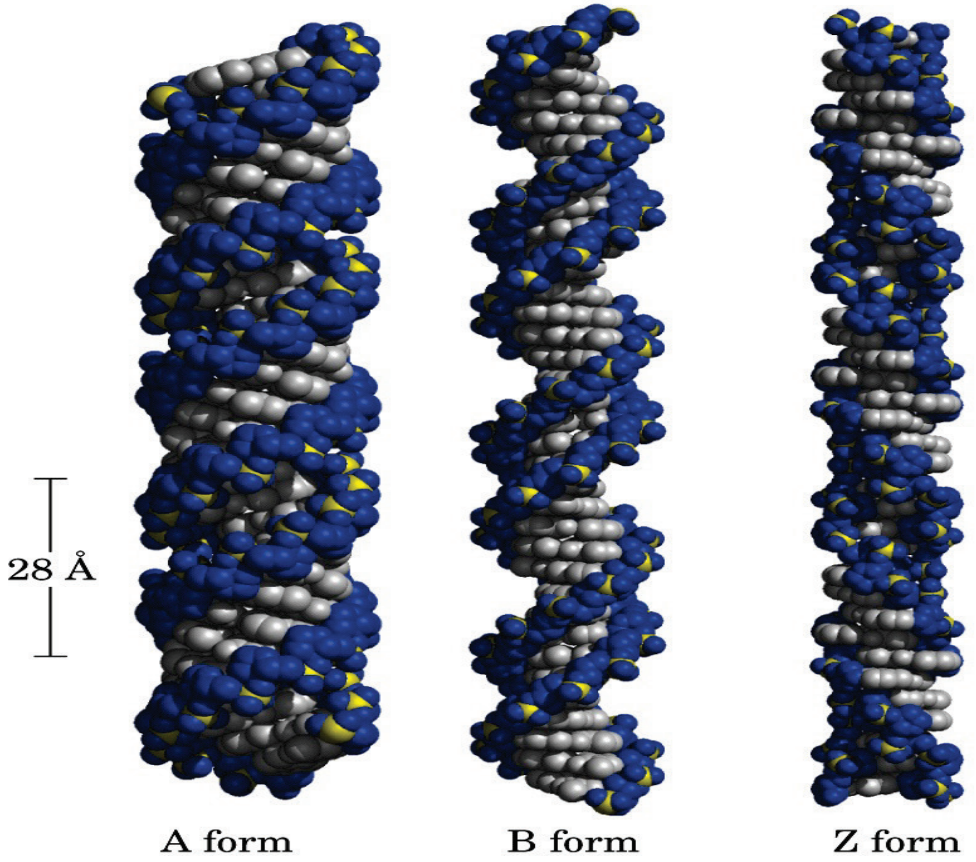
i) Conformations of the deoxyribose ring



DNA can adopt different conformations

Up to 11 different conformations have been described

	A form	B form	Z form
Helical sense	Right handed	Right handed	Left handed
Diameter	~26 Å	~20 Å	~18 Å
Base pairs per helical turn	11	10.5	12
Helix rise per base pair	2.6 Å	3.4 Å	3.7 Å
Base tilt normal to the helix axis	20°	6°	7°
Sugar pucker conformation	C-3′ endo	C-2′ endo	C-2′ endo for pyrimidines; C-3′ endo for purines
Glycosyl bond conformation	Anti	Anti	Anti for pyrimidines; syn for purines



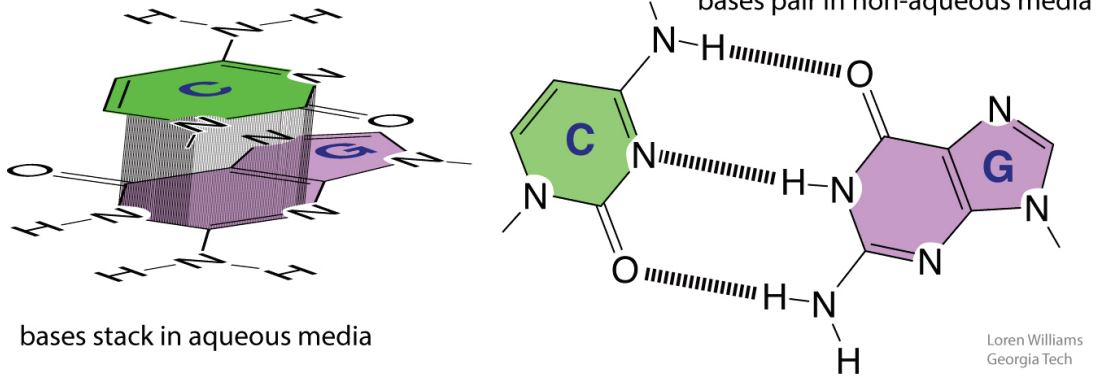
A Form: obtained in low humidity and high salt concentration conditions.
B Form: obtaned in high humidity and low salt concentrations conditions.
Z Form: obtained in high salt concentrations conditions.

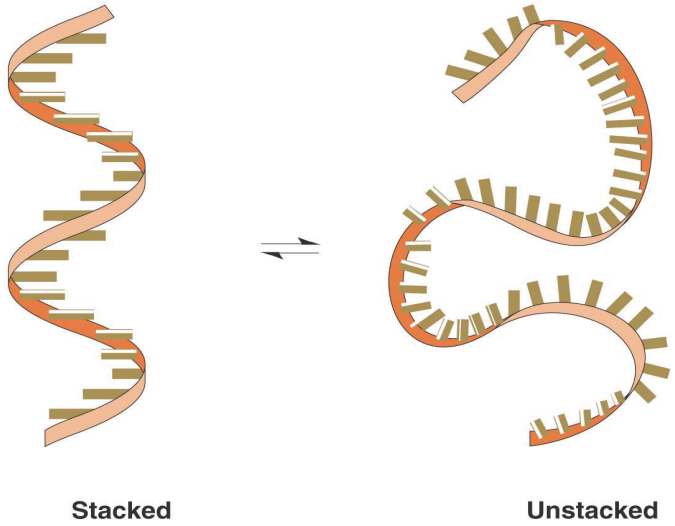
FACTORS THAT INFLUENCE THE STABILITY OF DNA DOUBLE HELIX

- Interactions between stacked nucleobases: a mix of hydrophobic and Van der Waals forces which are favored by the formation of the double hélix.

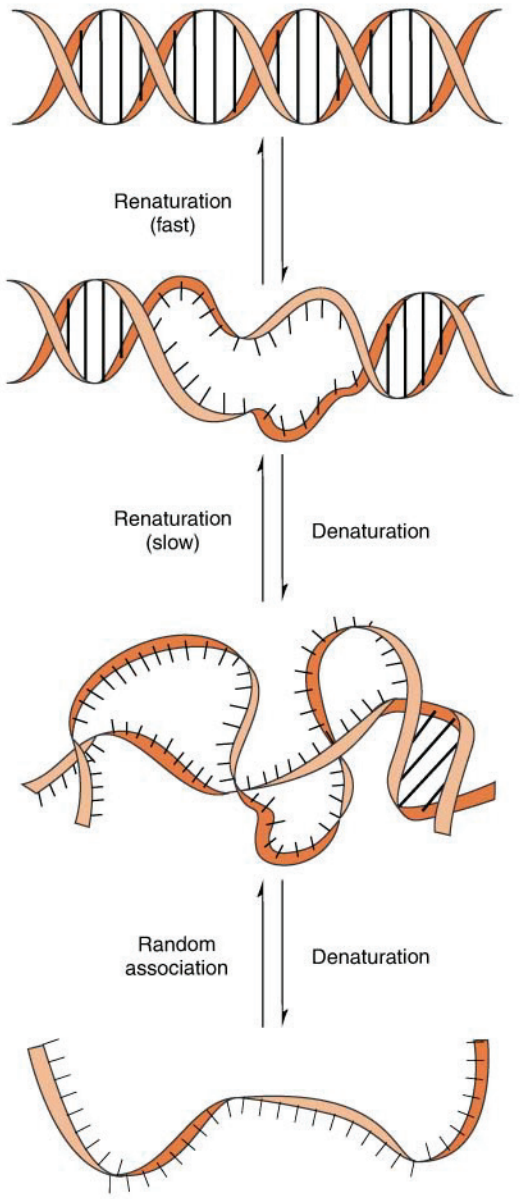
- The formation of numerous hydrogen bonds between complementary nucleobases on each strand. Energetically, these interactions are less important in the stabilization of the double helix, however they contribute to the correct orientation of the nucleobases facilitating stacking interactions.

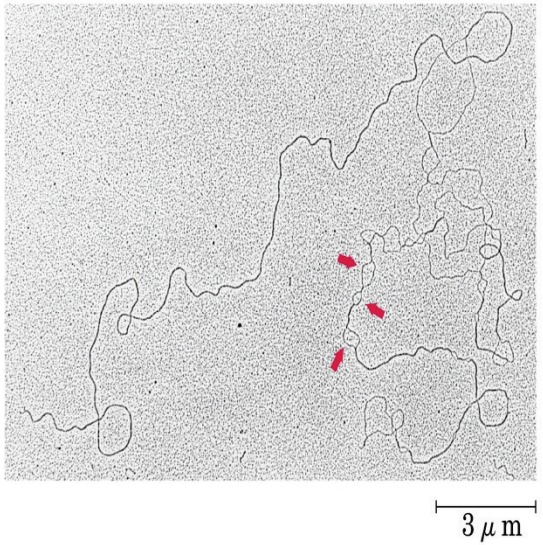
- Phosphate groups are negatively charged at physiological pH. The electrostatic interactions between phosphate groups from complementary DNA strands tend to destabilize the double helix. *In vivo* these negative charges are neutralized to a significant extent by their interactions with cations (such as $Mg^{2+,}$ polyamines and positively charged histone aminoacids).

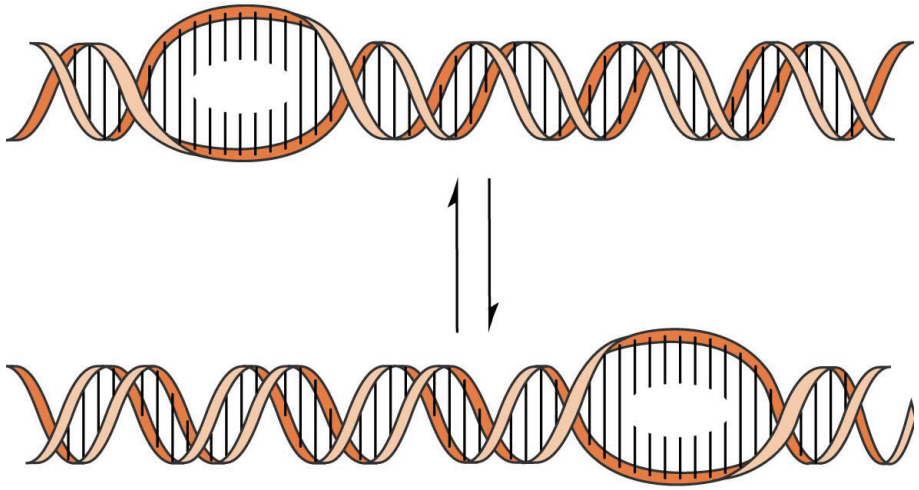




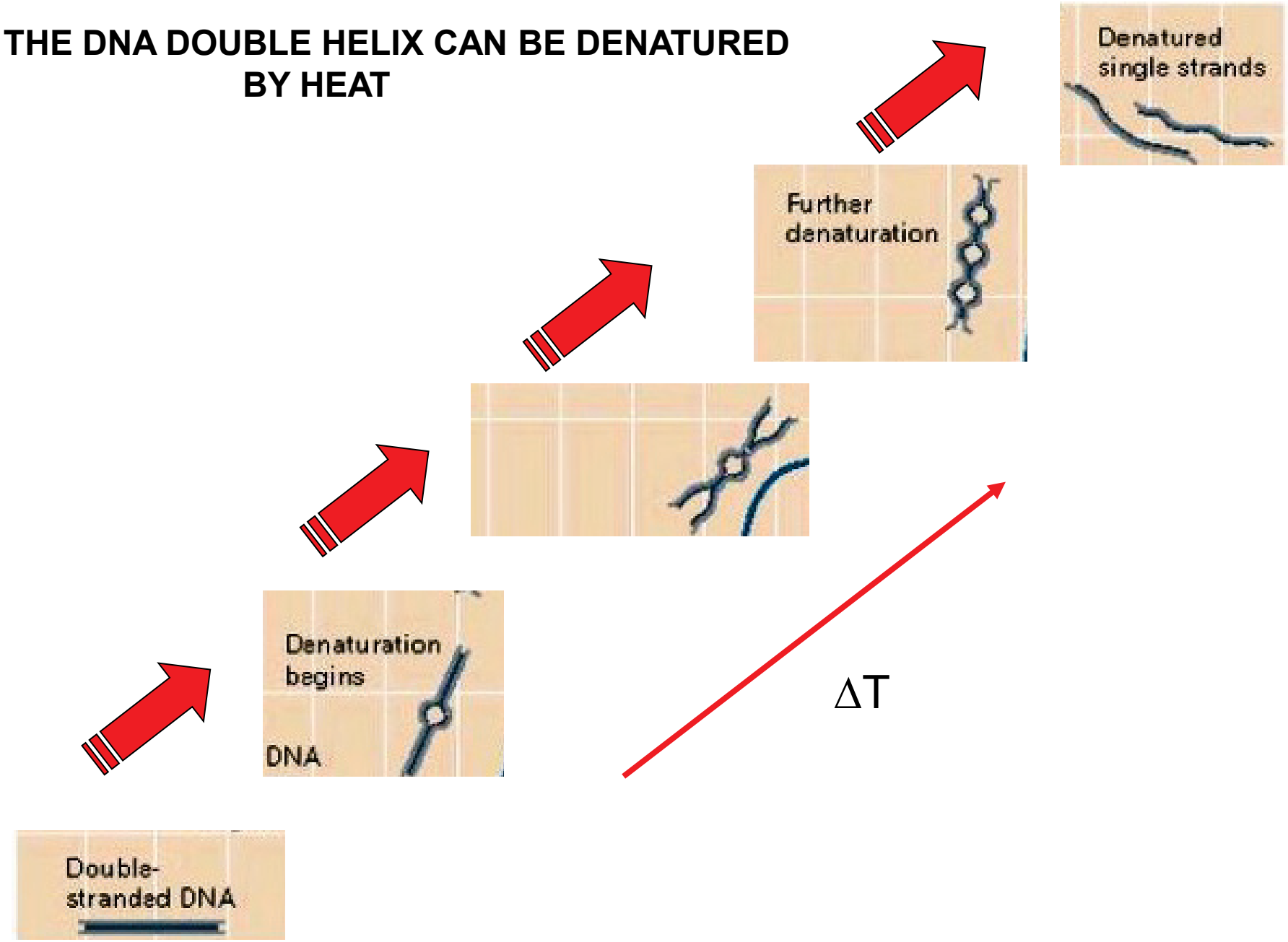
DENATURATION AND RENATURATION OF THE DNA DOUBLE HELIX



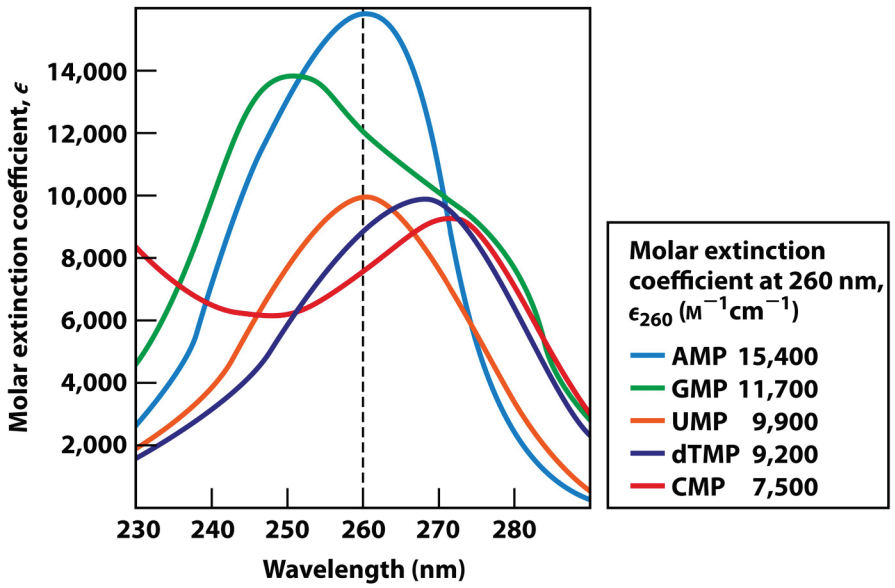




IN VITRO THE DNA DOUBLE HELIX CAN BE DENATURED BY HEAT

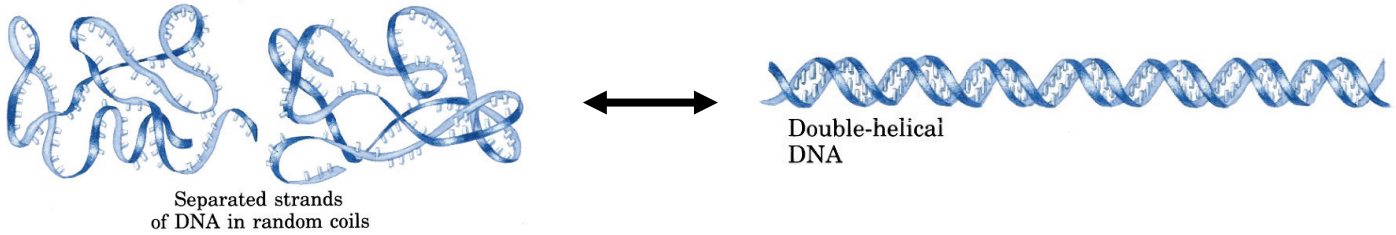


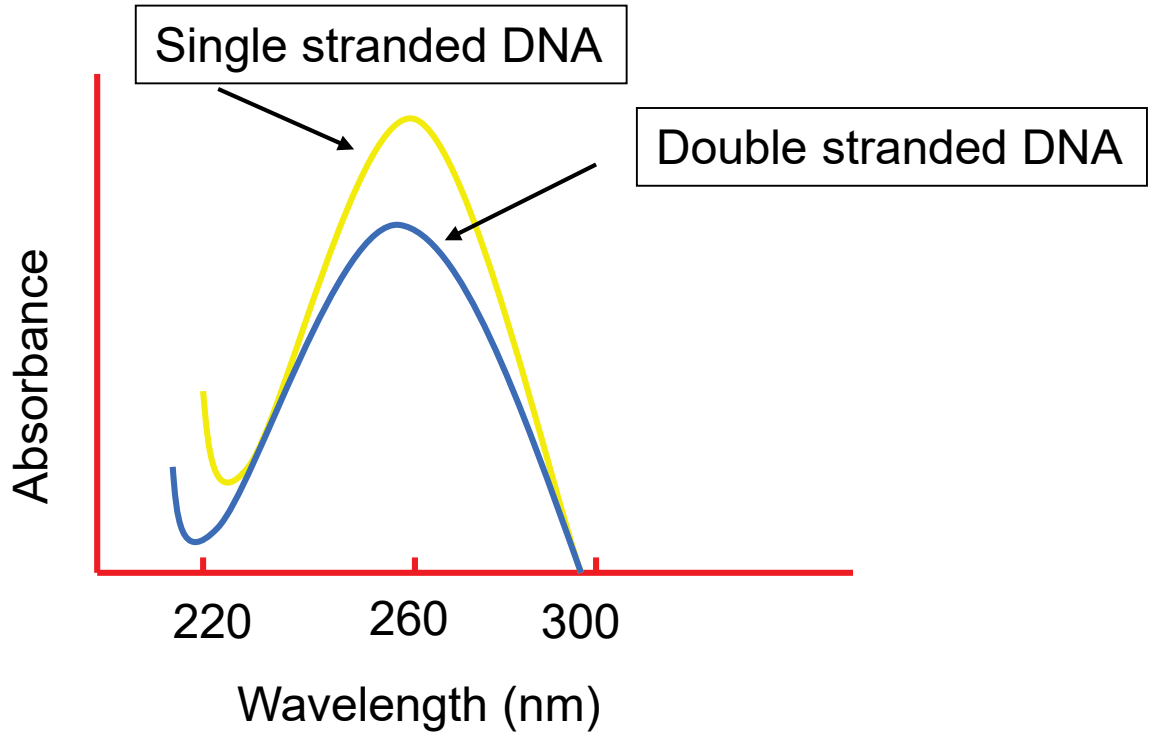
Nucleotides and nucleic acids absorb ultraviolet (UV) light



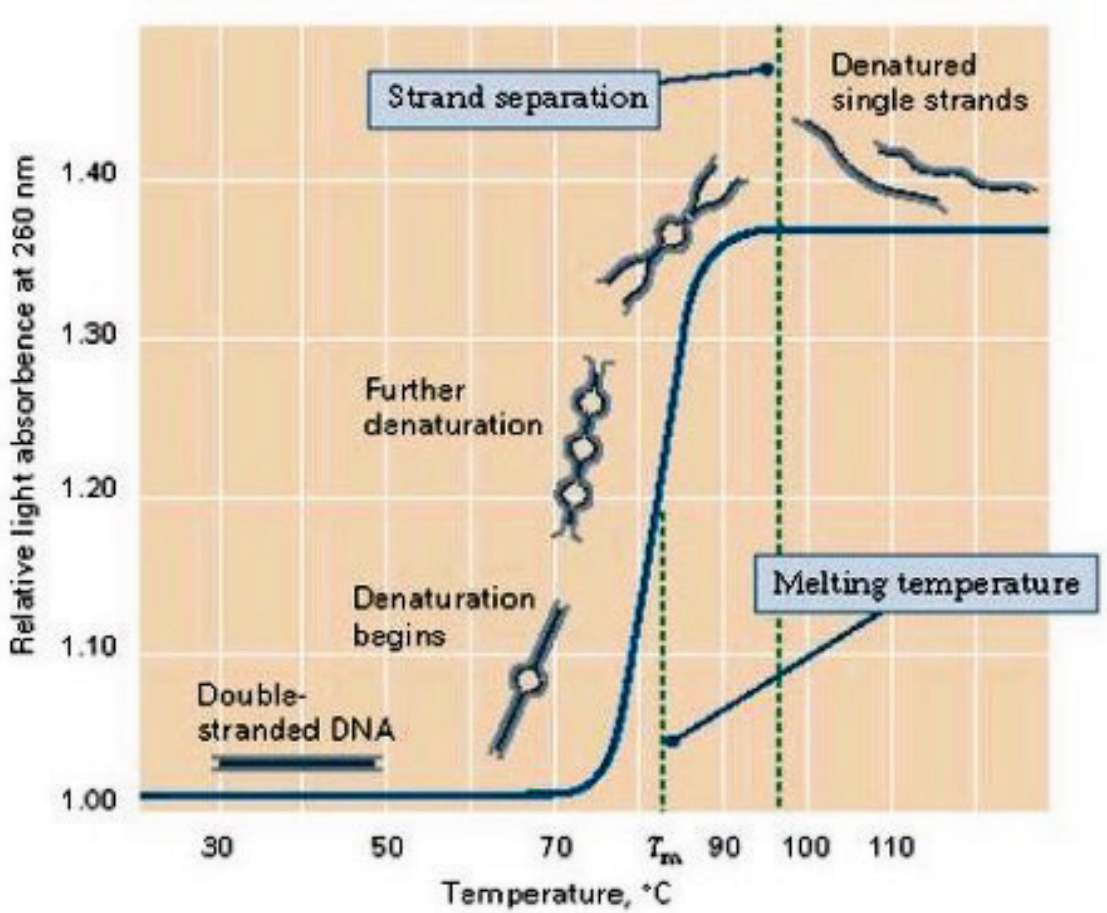
Double stranded DNA absorbs less UV light tan single stranded DNA

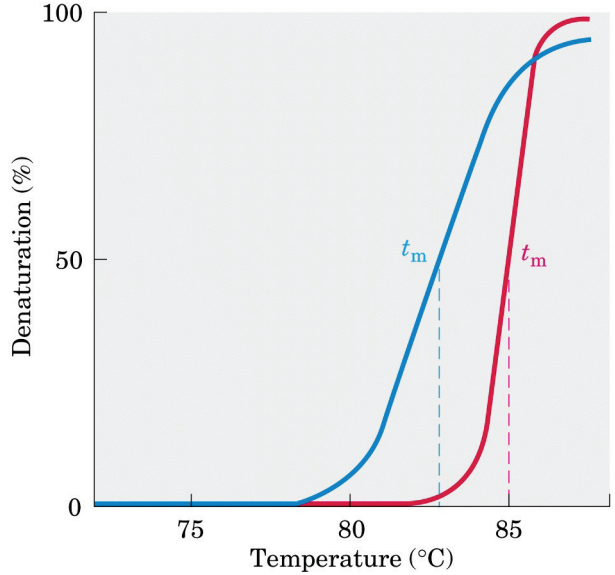
HYPOCHROMICITY

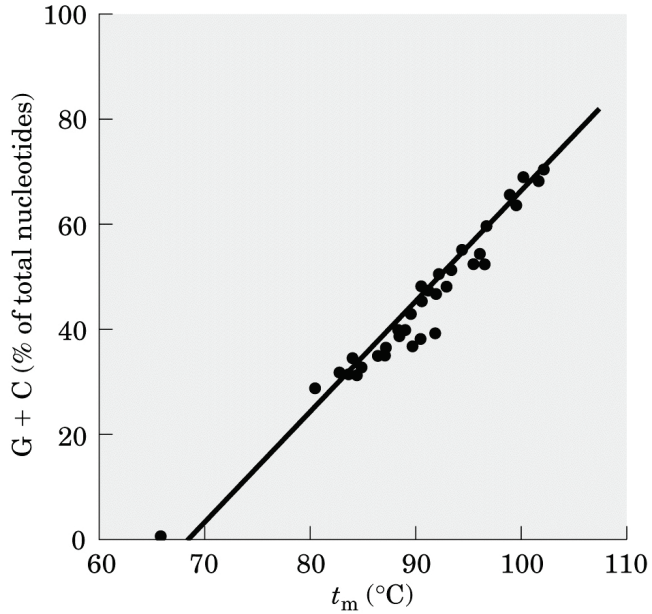




DNA denaturation can be followed by measuring light absorbance at 260 nm at increasing temperatures. The **“melting temperature”** (**Tm**) is that at which half of the double stranded structure is lost.







HYBRIDIZATION

DNA denaturation/renaturation and base-pair complementarity/hybridization are essential features of the polymerase-chain reaction (**PCR**) technology

